



Pricing and constructing international government bond portfolios

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ABSTRACT

This paper derives a stochastic discount factor for currency-hedged government bonds of developed markets by projecting returns onto the unconditional mean–variance efficient (UMVE) portfolio. Priced risks of international bonds differ fundamentally from those of currencies. The UMVE portfolio achieves a Sharpe ratio over twice the average of individual markets, with the market price of risk peaking during crises and periods with high inflation dispersion. While bond returns exhibit a strong factor structure, common sources of variation are only weakly connected to priced risks. Hedging unpriced risks in naive or factor-based strategies significantly improves Sharpe ratios, even under portfolio weight constraints.

1. Introduction

Sovereign bond markets represent one of the largest and most liquid asset classes globally, second only to public equity markets. Yet, little existing academic research exists on how internationally diversified government bond portfolios are priced, their risk–return characteristics, or the efficiency of alternative portfolio construction and trading strategies. In fact, sovereign bond pricing models are almost exclusively developed in single-country settings. Similarly, the literature on active government bond portfolio management primarily focuses on optimal duration choices or relative mispricing within a fixed-income market, with little discussion of allocation across bond markets (see, for example, the relevant chapters in [Bodie et al. \(2024\)](#)).

In contrast to the single-country focus of academic research, the availability of globally diversified government bond funds and exchange-traded funds (ETFs) has increased over the past years and the demand for these products is on the rise.¹

In this paper, we depart from the traditional single-country focus and take an asset class perspective to investigate the risk factors driving multi-country bond portfolios. We hereby assume that the positions in the different bond markets are financed in the respective local currencies, i.e. via short positions in the local money markets. Since this strategy is equivalent to the widely observed practice of managing foreign exchange (FX) risk by hedging via a FX-forward contract, we

will refer to these portfolios as *FX-risk-hedged*.² We derive several key results. To our knowledge, we are first to provide stylized facts on internationally diversified developed markets government bond portfolios. Our main sample comprises ten major developed sovereign debt markets, denoted as G10*, collectively representing approximately 70% of the global market capitalization for local-currency debt. In particular, we show that substantial diversification benefits arise even from naive diversification strategies, such as GDP-weighted portfolios, simply due to imperfectly correlated term-structure movements in the different markets.

Second, we construct the unconditional mean–variance efficient (UMVE) portfolio from the G10* markets, thus providing a stochastic discount factor (SDF). We find that the UMVE portfolio can be obtained using expected return forecasts based on forward spreads and real yields and a time-varying covariance matrix, estimated with a shrinkage method. The resulting SDF successfully prices all individual G10* bond markets as well as a range of dynamic trading strategies.

Third, we analyze the properties of the UMVE portfolio. We find that switching from single-country bond markets to the UMVE portfolio improves the Sharpe ratio substantially, from an average of 0.46 for individual markets to a value that exceeds 1. We also show that the expected Sharpe ratio of the UMVE exhibits substantial time-variation. Specifically, it increases around the Global Financial Crisis (GFC) in

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¹ The ratio of total net assets of global versus domestic fixed-income funds in the U.S. has approximately doubled over the past three decades. Using data from Morningstar, Figure A1 in the appendix plots this ratio and shows this ratio increasing from about 7.5% in 1995 to almost 15% in 2022. This number likely underestimates the actual international diversification of U.S. bond investors, as they may also achieve diversification by combining domestic bond funds from different countries within their portfolios.

² Throughout this paper, we use the terms “FX-risk-hedged” and “currency-hedged” interchangeably. The precise definition of bond excess returns is provided in Section 3.1.

2008, during the European sovereign debt crisis between 2010 and 2012 and during the COVID-19 crisis. The highest expected Sharpe ratio is observed in the fourth quarter of 2022, a period marked by inflation shocks, tightening monetary policies, and declining asset market valuations, which likely increased investors' marginal utility of wealth. Furthermore, we find that the expected Sharpe ratio of the UMVE portfolio is significantly related to inflation and that this relationship changes over time. Before the GFC, higher average trend inflation is associated with lower expected Sharpe ratios whereas the opposite is the case afterwards. In the post-GFC sample, we also observe a strong positive relation between the dispersion of inflation rates across markets and the UMVE's expected Sharpe ratio.

Fourth, the analysis reveals that there is very little relation between common sources of variation in international bond market returns and priced risks. A principal component analysis (PCA) shows that the G10* government bond returns exhibit a strong factor structure, with the first 3 principal components (PCs) explaining 86% of the total variance. However, these common sources of variation represent mostly unpriced risks, since all PCs combined account for only about 30% of the variation in UMVE returns.

We demonstrate that the UMVE portfolio's exposure to FX risk is economically negligible. Consistent with this, the UMVE portfolio returns are largely uncorrelated with an SDF that prices currencies. Notably, the component of the UMVE return that is orthogonal to the currency SDF prices international bond portfolios as well as the original UMVE portfolio. These results indicate that the priced risks of international bond portfolios are fundamentally distinct from those of currency portfolios.

Finally, we offer valuable insights for bond investors. Our analysis reveals that strategies such as naive diversification or factor-based approaches like carry, value, and momentum exhibit large amounts of unpriced risk. By constructing portfolios that hedge out these unpriced components, Sharpe ratios improve markedly. For instance, applying such a hedging strategy to a naively diversified (1/*N*) portfolio nearly doubles its Sharpe ratio. Substantial improvements are observed for other popular bond portfolio strategies, such as carry or value. Moreover, we find that, for practical implementation, accurately estimating expected returns is crucial, while estimation of variances plays a comparatively lesser role.

Implementing bond portfolio strategies that hedge unpriced risks often involves large absolute portfolio weights across different markets. To assess the feasibility of such strategies, we conduct sensitivity analyses and impose restrictions on the absolute values of portfolio weights. Our results indicate that portfolio performance remains remarkably robust under such constraints. Significant drops in performance are observed only when long-only restrictions are applied. However, such constraints appear excessively restrictive, as bond futures are available in all but one market in the sample, allowing professional market participants to easily short specific markets.

Our paper contributes to several strands of literature. First, it relates to research on return prediction in single domestic government bond markets. This literature has grown substantially since Fama and Bliss (1987), who document the empirical failure of the expectations hypothesis, showing that returns can be predicted using maturity-specific forward spreads. Cochrane and Piazzesi (2005) make fuller use of the forward curve for return prediction. They provide evidence that a common risk factor prices bonds of various maturities. Ludvigson and Ng (2009) show that macroeconomic and financial variables substantially improve the predictability of bond risk premia. Similarly, Cieslak and Povala (2015) highlight the critical role of inflation in predicting government bond returns, showing that deviations of bond yields from trend inflation carry valuable information about future returns.

Second, a limited number of papers examine international bond return predictability and yield curve fluctuation across different markets (e.g., Jotikasthira et al., 2015; Bekaert and Ermolov, 2023). Ilmanen (1995) analyzes bond predictability in six countries, finding that

expected bond returns tend to be high when the term spread and the real bond yield are elevated. Dahlquist and Hasseltoft (2013) extend the Cochrane–Piazzesi framework by constructing local and global factors, demonstrating that both predict international bond returns well. Brooks and Moskowitz (2019) instead find that carry and value predict excess returns on government bonds. Furthermore, Lustig et al. (2019) demonstrate that the predictability of unhedged foreign bond returns of developed countries decreases as their maturity increases. They attribute this to the offsetting movements of local currency term premia and currency premia at the long end of the yield curve.

Third, our paper contributes to the literature on SDFs derived from mean–variance efficient portfolios, as developed by Hansen and Richard (1987), Ferson and Siegel (2001), Ferson and Siegel (2003), Ferson and Siegel (2009), and, more recently, Chernov et al. (2022), Maurer et al. (2022) and Chernov et al. (2023). Our analysis also derives the unconditional mean–variance efficient portfolio, but we analyze a previously unexplored yet important class of assets: international government bonds.

In summary, the existing related literature primarily adopts a single-market perspective on government bonds or focuses on entirely other asset classes, such as the stock market or currency markets. Our approach differs, as we analyze the pricing of developed countries' government bonds as a distinct asset class. We derive the UMVE portfolio out-of-sample, provide insights about its ability to price bond markets and strategies, about its average realized Sharpe ratio, the time series properties of the resulting expected risk premia, and on how common sources of return variation in international bond portfolios are related to priced risks.

The paper proceeds as follows. Section 2 outlines the theoretical foundations. Section 3 describes the empirical strategy and presents stylized evidence. Section 4 presents the main empirical findings. The role of exchange rates is analyzed in Section 5. Section 6 explores the implications for bond portfolio strategies, and Section 7 provides robustness checks. Section 8 concludes.

2. A linear factor model

This section derives the theoretical foundations for the empirical strategy that will be implemented in the remainder of the paper. Specifically, we derive an SDF that prices international bond markets as well as dynamic trading strategies both conditionally and unconditionally via the UMVE portfolio. The foundations for this approach have been developed by Hansen and Richard (1987), Ferson and Siegel (2001), and, more recently, by Chernov et al. (2022). The following presentation builds on this literature.

The UMVE portfolio is defined by a dynamic trading strategy in the *N* assets with excess returns $rx_{t+1}^U = (w_t^U)^T rx_{t+1}$. Its weights w_t^U are a function of the time-*t* covariance matrix $\nabla_t(rx_{t+1})$ and expected excess returns $\mathbb{E}_t(rx_{t+1})$:

$$w_t^U = \frac{\nabla_t^{-1}(rx_{t+1})\mathbb{E}_t(rx_{t+1})}{1 + \mathbb{E}_t(rx_{t+1})^T \nabla_t^{-1}(rx_{t+1})\mathbb{E}_t(rx_{t+1})}. \quad (1)$$

While a portfolio with weights proportional to $\nabla_t^{-1}(rx_{t+1})\mathbb{E}_t(rx_{t+1})$ is always conditionally mean–variance efficient, this is not true unconditionally. Obtaining the UMVE portfolio requires scaling the portfolio weights by $(1 + \mathbb{E}_t(rx_{t+1})^T \nabla_t^{-1}(rx_{t+1})\mathbb{E}_t(rx_{t+1}))^{-1}$. As has been shown by Hansen and Richard (1987), the unconditionally efficient portfolio must be conditionally efficient, while the converse is not true.

The UMVE prices all individual assets and any combination of them, including dynamic trading strategies. Specifically, we can write both conditional and unconditional linear factor pricing models in terms of the UMVE. The unconditional relationship for any portfolio *p* is:

$$\mathbb{E}(rx_{t+1}^p) = \beta^p \mathbb{E}(rx_{t+1}^U), \quad (2)$$

where the unconditional beta $\beta^p = Cov(rx_{t+1}^p, rx_{t+1}^U) \nabla^{-1}(rx_{t+1}^U)$. This relationship holds for any asset or portfolio strategy *p*.

Similarly, the conditional version of the asset pricing model is given by

$$\mathbb{E}_t(rx_{t+1}^p) = \beta_t^p \mathbb{E}_t(rx_{t+1}^U), \quad (3)$$

with a conditional beta $\beta_t^p = Cov_t(rx_{t+1}^p, rx_{t+1}^U) \mathbb{V}_t^{-1}(rx_{t+1}^U)$.

Finally, the UMVE-implied SDF is

$$M_{t+1} = 1 - (rx_{t+1}^U - \mathbb{E}(rx_{t+1}^U)), \quad (4)$$

and it satisfies

$$\mathbb{E}_t(M_{t+1} rx_{t+1}^p) = 0 \quad (5)$$

for all portfolios and dynamic trading strategies p in the basis assets. It is important to point out that this SDF M_{t+1} , defined in Eq. (4), is estimated from the UMVE portfolio returns. Thus, it can be interpreted as an unconditional linear projection of the true SDF onto the payoff space of all possible trading strategies that can be constructed from the N basis assets.

3. Empirical strategy and stylized facts

Empirical implementation of the approach described in Section 2 to construct the UMVE portfolio requires the estimation of (i) expected returns and (ii) the conditional variance. To determine the weights of the UMVE portfolio at any time t , we will use only information available at that point in time, while returns are observed out-of-sample, in the subsequent period.

The estimation of conditional covariances faces substantial challenges in a high-dimensional asset space, such as in the equity market. Indeed, the estimated covariance matrix might not have full rank or even exhibit negative eigenvalues. Therefore, the low-dimensional asset space resulting from our focus on a small cross-section of 10 major government bond markets provides an important advantage when estimating the variance–covariance matrix.

The remainder of this section explains how bond excess returns are defined and presents stylized evidence for the markets in our sample. Furthermore, it describes how expected excess returns and their conditional covariances are estimated and reports the data sources.

3.1. Bond excess returns

To analyze the pricing of internationally diversified bond portfolios, we consider a portfolio strategy where each month investments in local currency bonds are financed via short-term local-currency loans. Each country's bond market is represented by a synthetic 10-year zero-coupon bond, obtained from the appropriate zero-coupon bond curve.³ The corresponding holding period excess return, rx_{t+1}^i , is therefore given by

$$rx_{t+1}^i := rx_{i,t \rightarrow t+1M}^{(10Y)} = \left(\frac{P_{i,t+1M}^{(10Y-1M)}}{P_{i,t}^{(10Y)}} - \frac{1}{P_{i,t}^{(1M)}} \right) \frac{S_{i,t+1M}}{S_{i,t}}, \quad (6)$$

where $P_{i,t}^{(10Y)}$ and $P_{i,t}^{(1M)}$ are the local currency prices of a 10-year government bond and of a 1-month government bill at time t for country i , respectively, and $S_{i,t}$ is the exchange rate of country i , expressed as the price of the foreign currency in units of the home currency, which we interpret to be the U.S. dollar. Time series of excess returns, as defined in Eq. (6), are calculated at monthly frequency, using data from the last day of each month. The expression in Eq. (6) defines an FX-risk-hedged bond excess return, as it results from a long position in the foreign currency bond that is financed by a short position in a 1-month local currency money market instrument. Equivalently, it can be interpreted

as a long position in the foreign currency bond that is funded via a short position in a 1-month money market instrument in the home currency and an appropriate short position in a 1-month currency forward. In contrast, an outright position in the foreign currency bond would be fully exposed to FX risk, as the second term in the bracket would not be subtracted.⁴

The return definition in Eq. (6) matches the one used in other studies, such as in Koijen et al. (2018), which the authors refer to as “hedged dollar return”. While this form of FX management is common among bond investors, it does not completely eliminate FX risk, as is apparent from Eq. (6).⁵ Section 5 therefore provides a detailed analysis of the remaining FX risk in bond returns defined in Eq. (6) and to what extent any remaining FX risk is priced.

In our main empirical analysis we focus on government bond markets of the following countries, which we refer to as G10*: Australia, Canada, Germany, Japan, New Zealand, Norway, Sweden, Switzerland, the United Kingdom, and the United States. The currencies of these countries are the ones which are frequently considered in the foreign exchange literature, which refers to them as “G10 currencies” (see, for example, Lustig et al. (2019) and Chernov et al. (2023)).

Government bonds of these countries are typically issued in local markets and denominated in local currencies (Chen et al., 2019). In fact, 99.8% of the outstanding long-term central government debt of the countries in our main sample is denominated in domestic currency, according to the Bank for International Settlements (BIS).⁶ The analysis in this study is therefore based on local currency bonds. The Eurozone is represented by German government bonds, arguably the safest and most liquid ones.

The countries in the main sample represent about 70% of the global local-currency long-term debt in the BIS statistic, as of the end of 2022. As we use only German bonds to represent the Eurozone, and since Euro area government bonds tend to be highly correlated, our implicit coverage of world government bond markets is plausibly even larger. Finally, the issuing countries in the sample are very unlikely to default, such that their government bonds can be considered as practically default-risk free assets.

3.2. Data

To calculate bond excess returns, as defined in Eq. (6), we collect daily data from Bloomberg on long-term and short-term zero-coupon yields for the period from January 1995 to December 2022 for the G10* countries.⁷ For some analyses, we expand the set of countries and thus we also obtain data on zero-coupon yields and exchange rates for additional markets. For this extended sample, we select all OECD member countries which do not belong to the G10* and for which data availability for the zero-yield curve from Bloomberg is adequate. This process defines a sample of 20 countries. In addition we consider Hong Kong and Singapore for which sufficiently long time series of zero yield curves are also available and which have liquid and sizable government

⁴ For a detailed exposition of the equivalence of the returns defined above and FX forward-hedged returns, we refer to Appendix B.

⁵ Campbell et al. (2010) find that the risk-minimizing strategy for a global bond investor is indeed close to a full currency hedge, which corresponds to return defined in Eq. (6). Consistent with this finding, Sialm and Zhu (2024) provide empirical evidence that around 90% of U.S. international fixed-income mutual funds use currency forwards to manage their foreign exchange exposure (although they tend to hedge only partially). They also show that hedging foreign exchange risk does not decrease the funds' abnormal returns, but reduces their return variability.

⁶ Data source: BIS, <http://www.bis.org/statistics/c4.pdf>. The percentage does not include New Zealand, where data are unavailable in the BIS statistic.

⁷ Since Bloomberg does not provide data on the 1-month yield for all countries in our sample, we approximate it using the 3-month yield, which is consistently available.

³ This maturity is frequently used by practitioners and in the academic literature to represent government bond markets (see, e.g., Jordà et al., 2019; Lustig et al., 2019).

Table 1
Descriptive statistics of bond returns.

	Mean	Median	Std. Dev.	SR
Australia	3.906	3.501	9.007	0.434
Canada	4.364	3.240	7.498	0.582
Germany	4.105	6.814	6.861	0.598
Japan	3.646	2.796	5.103	0.715
New Zealand	2.791	4.571	8.515	0.328
Norway	3.226	3.135	7.352	0.439
Sweden	5.313	5.612	8.100	0.656
Switzerland	3.486	4.521	5.717	0.610
United Kingdom	3.401	3.158	8.136	0.418
United States	4.310	2.214	9.013	0.478

This table shows descriptive statistics of monthly 10-year government bond excess returns (in % p.a.) and annualized Sharpe ratios (SR) by country of issuance. Statistics refer to the G10* countries. Bond returns are in monthly frequency. Data are from January 1995 to December 2022 and measured in USD.

bond and foreign exchange markets. The final list of extended countries is composed of: Austria, Belgium, Finland, France, Greece, Ireland, Italy, the Netherlands, Portugal, Slovakia, Slovenia and Spain, which belong to the Euro area as of 2022, and Chile, Denmark, Colombia, Czech Republic, Hong Kong, Hungary, Israel, Mexico, Singapore, and Turkey, which are countries outside the Euro area.

We also use several macro data sources. From Datastream, we collect monthly data on consumer price indices (CPIs), unemployment rates, and quarterly data on gross domestic product (GDP) for each country in our sample.⁸ We then compute the cross-sectional standard deviation of these variables as a measure of dispersion across macroeconomic dimensions. As proxies for the return and volatility of the stock market, we collect data for the MSCI World Index and for the VIX Index from Bloomberg. The Global Economic Policy Uncertainty Index (EPU) is obtained from the website of Scott Baker, Nick Bloom, and Steven Davis. This index is based on Baker et al. (2016) and measures global policy-related economic uncertainty. We also obtain data on the intermediary capital risk factor, defined by He et al. (2017) as AR(1) innovations to the capital ratio, scaled by its lagged value, where the capital ratio is defined as the aggregate value of market equity divided by the aggregate value of total assets (i.e., the sum of aggregate market equity and book debt) and computed for the largest financial intermediaries in the U.S. These data are available from the authors' websites.⁹

3.3. Stylized facts

We first provide empirical facts about the risk-return characteristics of government bond markets and dynamic government bond portfolio strategies. Using Eq. (6), we calculate annualized average excess returns, standard deviations, and Sharpe ratios for the G10* countries. Table 1 provides descriptive statistics for annualized one-month holding period excess returns of 10-year government bonds by country of issuance. Although all countries can be considered as very close to default free, investors earned attractive returns over the sample period. In fact, Sharpe ratios are above 0.4 in all markets except New Zealand. The high average returns over the sample period partially reflect a worldwide downwards trend in bond yields, which persisted until March 2022.

Table 2 compares risk-return characteristics from individual bond markets with those that can be obtained from naively diversified static portfolios and from simple dynamic portfolio strategies. Comparing

⁸ For the GDP data, to move from quarterly to monthly frequency, we assign the same value for all three months of each quarter.

⁹ Table C2 in the appendix provides descriptive statistics for the above macro and market variables.

Table 2
Stylized evidence.

	Mean	Median	Std. Dev.	SR
<i>A. No diversification</i>				
Individual countries' average	3.307	3.145	7.379	0.463
<i>B. Simple strategies</i>				
Naive (1/N)	3.307	2.309	6.077	0.544
GDP-weighted	3.505	2.862	6.637	0.528
<i>C. Conditional strategies</i>				
Carry	3.262	2.684	4.011	0.813
Value	1.815	1.265	3.907	0.465
Momentum	1.098	0.961	3.860	0.284

This table shows the mean return in % p.a., median return in % p.a., standard deviation in % p.a., and annualized Sharpe ratio (SR) of individual countries' average as well as global bond portfolio strategies. The sample consists of Australia, Canada, Germany, Japan, New Zealand, Norway, Sweden, Switzerland, the United Kingdom, and the United States. Panel A shows results with no diversification. Panel B shows results of simple portfolio strategies. Panel C presents conditional strategies, where portfolios are constructed using rank-weights following Asness et al. (2013) and focusing on the following signals respectively: bond carry, defined as the difference between 10-year yield and the local 1-month yield; bond value, defined as the difference between the 10-year yield and inflation; and bond momentum, defined as the cumulative return over 12 months skipping the last month. Data are from December 1995 to December 2022.

Panel A of Table 2, which denotes the benchmark case with no diversification, to Panel B, which displays simple diversification strategies, it becomes clear that international diversification matters for bond investors. By definition, naive (1/N) diversification across markets achieves an average excess return identical to the cross-sectional average of the single-countries in the sample, which equals 3.3% p.a. The individual countries' average standard deviation equals 7.4%. By diversifying, investors could have lowered the standard deviation of their portfolio considerably by almost 20% to 6.1%, thereby achieving a substantially higher Sharpe ratio. While the individual countries' average Sharpe ratio equals 0.46 in our sample, naive diversification across the 10 markets increases the Sharpe ratio to 0.54. A GDP-weighted strategy yields an only marginally worse risk-return trade-off, with a Sharpe ratio of 0.53. Panel C of Table 2 explores whether simple portfolio strategies already realize most of the benefits of internationally diversified bond portfolios, or whether conditional strategies, which should be easily accessible to bond portfolio managers, lead to further improvements. To this end, we investigate long-short portfolios sorted on carry, value and momentum. Returns to these strategies have been shown to contribute significantly to portfolio performance in different asset classes, such as equity, currencies, and fixed income (e.g., Fama and French, 2012; Asness et al., 2013; Kojien et al., 2018). We therefore use the following characteristics to construct simple dynamic portfolio strategies: the slope of the yield curve, i.e. the difference between the 10-year yield and the local 1-month yield (*carry*), the real bond yield, i.e. the difference between the 10-year yield and inflation (*value*), and momentum, i.e. the cumulative return over 12 months skipping the most recent one (*momentum*). We document that all long-short bond portfolios achieve positive performance. The Sharpe ratios for carry, value and momentum are 0.81, 0.47, and 0.28, respectively. Interestingly, the carry strategy outperforms all others substantially. This evidence is consistent with the literature highlighting that returns from carry generate positive alpha for a host of asset classes (Kojien et al., 2018).

Fig. 1 illustrates time variation in returns and risk for the bond portfolio strategies. Specifically, Panels A and B display the extent to which average excess returns fluctuate over time by plotting moving averages over 36 months for the returns of the naive and the GDP-weighted portfolios, as well as the factor portfolios carry, value, and momentum from Table 2 (Panel C). The considerable heterogeneity in the performance dynamics of these bond portfolio strategies further supports the conjecture that investors might be able to benefit from conditional strategies. Fig. 1, Panels C and D instead show the time

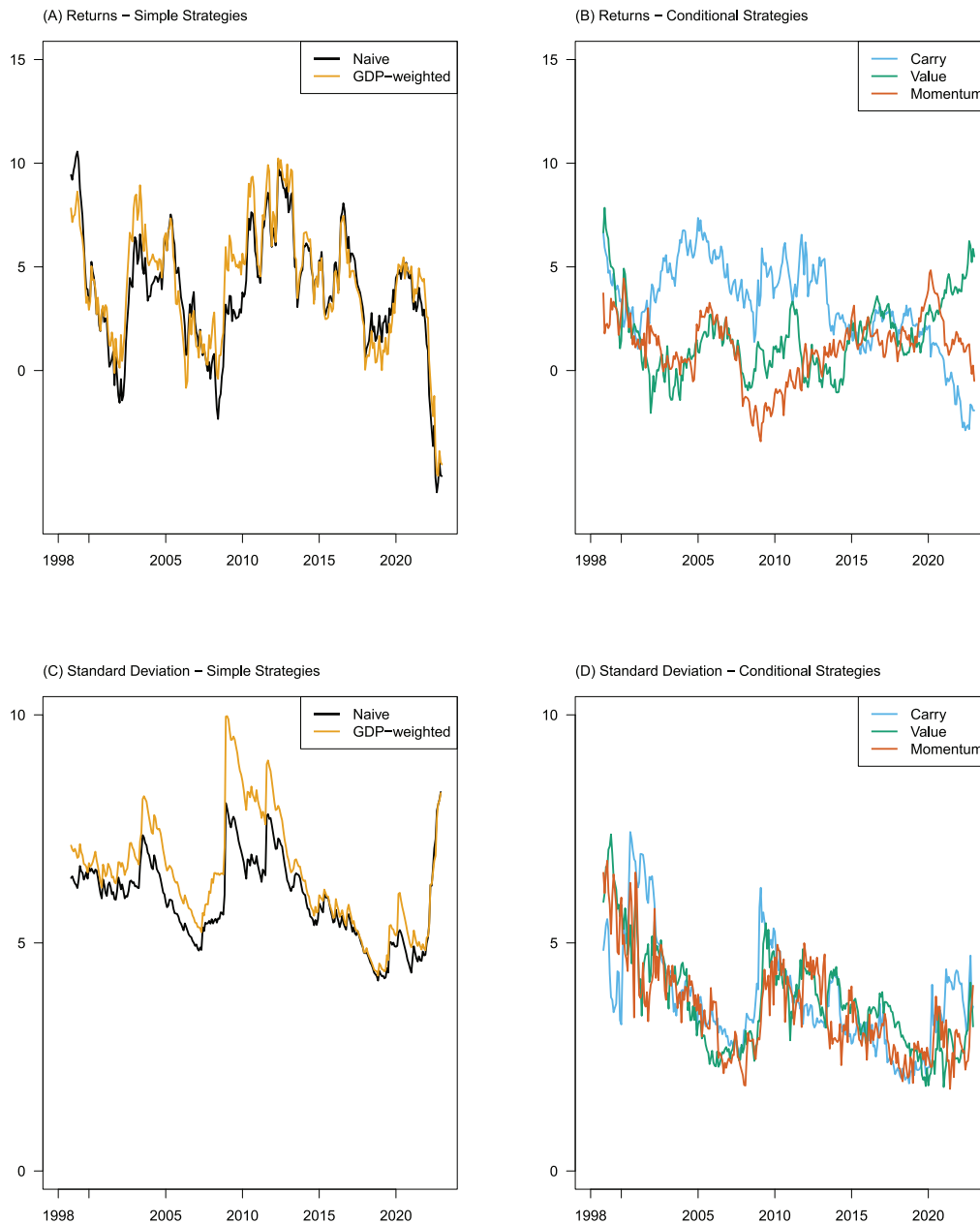


Fig. 1. Time variation in returns and standard deviations.

Panel A shows annualized moving average returns over 36 months for the naive ($1/N$) and the GDP-weighted portfolios. Panel B shows moving average returns over 36 months for the rank-weighted conditional portfolios (Asness et al., 2013). These are constructed using bond carry, value, and momentum as signals. Bond carry is defined as the difference between 10-year yield and the local 1-month yield. Bond value is defined as the difference between the 10-year yield and inflation. Bond momentum is defined as the cumulative return over 12 months skipping the last month. Panels C and D plot standard deviations for the simple and conditional strategies, respectively. Details on the covariance matrix estimation, based on Ledito and Wolf (2022), which is used for the standard deviations, are reported in Section 3.4.2. Bond data belong to the following countries: Australia, Canada, Germany, Japan, New Zealand, Norway, Sweden, Switzerland, United Kingdom, United States. Data are from October 1998 to December 2022.

series of the standard deviations of these five portfolios, where the estimation procedure used to compute the covariances is explained in detail in Section 3.4.2. While it can be seen that all portfolios show considerable fluctuations in their standard deviations over time, the factor strategies appear to exhibit lower risk levels on average. Motivated by these findings, we proceed to analyze the properties of the UMVE portfolio, using the parsimonious theoretical framework described in Section 2. To this end, we first explain how the two inputs for its estimation, namely expected returns and conditional covariance matrix are derived.

3.4. Estimation of expected bond returns and covariances

3.4.1. Expected returns

To forecast expected excess returns $\mathbb{E}_t(r_{t+1}^i)$ one month ahead, we rely on forward spread and value signals. These are two well established predictive variables, for which there is a large empirical literature, both as potential predictors of excess returns in local bond markets as well as of other asset classes. Specifically, we run the following pooled ordinary least squares (OLS) regression, using expanding time windows, with standard errors clustered at the country level:

$$\mathbb{E}_t(r_{t+1}^i) = \gamma_t \text{ForwardSpread}_{i,t} + \delta_t \text{Value}_{i,t}, \quad (7)$$

where we define the explanatory variables as follows. The forward spread is defined as

$$ForwardSpread_{i,t} = \frac{P_{i,t}^{(10Y-1M)}}{P_{i,t}^{(10Y)}} - \frac{1}{P_{i,t}^{(1M)}}. \quad (8)$$

Many papers, starting with Fama and Bliss (1987) and Campbell and Shiller (1991), document the empirical failure of the expectations hypothesis and that forward rates predict future bond returns: steep yield curves signal high subsequent holding period returns. More recently, Kojien et al. (2018) provide evidence that similar carry measures predict returns cross-sectionally and in the time series for different asset classes. Their definition of global bond carry is a slightly modified version of the Fama and Bliss (1987) forward spread. Linking carry concepts from the currency and bond markets, Andrews et al. (2024) show that sorting countries on the slope of their yield curves constitutes a profitable trading strategy, in particular since 2008 when returns from the traditional currency carry have weakened.

Next we define value by the real bond yield:

$$Value_{i,t} = Y_{i,t}^{(10Y)} - \pi_{i,t}, \quad (9)$$

where $\pi_{i,t} = CPI_{i,t}/CPI_{i,t-12M} - 1$ and $CPI_{i,t}$ is country i 's consumer price index, lagged by one month. Inflation has been considered an important driver of bond holding period returns, both in academic research as well as among practitioners. Campbell and Ammer (1993) show that unexpected excess bond returns must be associated with changes in expected inflation rates, future real interest rates, or future excess bond returns. Because bond payoffs are fixed, changes in expected inflation rates impact nominal bond returns even if expected real returns are constant. More recently, Cieslak and Povala (2015) find that a cycle factor that captures deviations of observed bond yields from those fitted on the basis of current and past inflation predicts excess returns of long-term yields.

It is plausible that sophisticated investors were aware of the relation between the above predictive variables and bond returns throughout our sample period. As discussed above, the shape of the yield curve has already been identified as a determinant of bond risk premia by Fama and Bliss (1987) and Campbell and Ammer (1993), who split the bond yield into expected real rates and other components, thus emphasizing the role of value measures as defined in Eq. (9) above. Also, Ilmanen (1995) considers both the term spread and the real bond yield as instruments to explain bond excess returns. To ensure that we use only information up to time t to form expectations about returns ending in period $t+1$, we re-estimate the coefficients in Eq. (7) every month using an expanding sample. For the regression, we require a minimum of 60 months of data, and thus we obtain expected returns starting with January 2001.

Fig. 2 shows the time series of the estimated coefficients $\hat{\gamma}_t$ and $\hat{\delta}_t$, while Table 3 reports their descriptive statistics. Even though there is some time variation in the estimates, they appear to be quite stable. Consistent with the literature, both the forward spread and bond value have positive coefficients, but only the former has, on average, a statistically significant relation with expected returns. In Table 3, the average t -statistics are 2.74 and 1.15 for the forward spread and bond value, respectively. Despite the low t -stat of value, it turns out to be an important predictor. As will be seen in Section 4.2, both the forward spread and value signals are required to obtain a UMVE portfolio that prices international bond portfolios.¹⁰

¹⁰ The importance of value as an additional regressor is also confirmed by the Diebold and Mariano (1995) test comparing the model that includes both signals with the one that only uses the forward spread. The test shows a significantly higher predictive ability for the full model, with a t -statistic of -3.45 . Furthermore, Table 8 shows that a restricted UMVE, which is constructed using only the forward spread, is not able to price dynamic portfolio strategies.

3.4.2. Covariances

We estimate the covariance matrix using a three-step procedure. First, we compute monthly bond returns at daily frequency. We require a high frequency of observations in this stage as this allows us to estimate a covariance matrix $\hat{\Sigma}_t$ for each month, thus capturing time variation. Second, given the low number of observations within a single month, we apply the quadratic shrinkage method of Ledoit and Wolf (2022), obtaining $\tilde{\Sigma}_t$. Finally, we calculate an exponentially weighted average of the shrunk covariance matrices, to arrive at the estimate of the covariance matrix $\mathbb{V}_t$:

$$\mathbb{V}_t = (1 - \lambda)\tilde{\Sigma}_t + \lambda\mathbb{V}_{t-1}, \quad (10)$$

where we set $\lambda=0.94$.¹¹

3.4.3. Unbiasedness

As a sanity test, we first check if the main objects that we use for constructing the UMVE portfolio are unbiased. In particular, we regress (i) realized excess returns rx_{t+1}^i on expected excess returns $\mathbb{E}_t(rx_{t+1}^i)$, (ii) squared returns in excess of expected values, $(rx_{t+1}^i - \mathbb{E}_t(rx_{t+1}^i))^2$, on estimated variances, $\sigma_t^2(rx_{t+1}^i)$, which are the diagonal elements of the matrix $\mathbb{V}_t$, and (iii) products of returns $(rx_{t+1}^i - \mathbb{E}_t(rx_{t+1}^i)) \cdot (rx_{t+1}^j - \mathbb{E}_t(rx_{t+1}^j))$ on estimated covariances $cov_t(rx_{t+1}^i, rx_{t+1}^j)$, which are the off-diagonal elements of $\mathbb{V}_t$.

Table 4 shows that the regression of realized returns on predicted returns produces a coefficient close to 1 (the estimated coefficient is equal to 0.91). The conditional variance also appears consistently estimated, as indicated by the coefficient of 0.95. Similar conclusions obtain for the estimated covariance, where the regression coefficient almost exactly equals 1.00. Thus, according to Table 4, the predictive regressions are unbiased, as in none of the cases the coefficients are statistically significantly different from 1.¹²

4. Empirical evidence

After defining the set of the test assets, this section provides the main results of the asset pricing tests to evaluate the linear factor model, discusses the implications, and analyzes the properties of the UMVE portfolio returns.

4.1. Test assets

Following Hansen and Richard (1987), one can evaluate the conditional linear beta pricing relationship from Eq. (3) by testing the unconditional linear beta pricing relation specified in Eq. (2) for a comprehensive set of dynamic trading strategies as test assets. Therefore, in addition to individual countries' bond returns, we include 14 trading strategies in our set of test assets. Specifically, we define the return of a dynamic portfolio strategy P , rx_{t+1}^P , as

$$rx_{t+1}^P = \sum_{i=1}^{N_t} w_{i,t}^P rx_{t+1}^i, \quad (11)$$

where $w_{i,t}^P$ is the weight of country i in strategy P and N_t is the number of countries at time t . In addition to the naive $1/N$ and the GDP-weighted portfolios, we use bond carry, value, momentum, and short-term momentum as trading signals, which we define as follows:

¹¹ We set $\mathbb{V}_1$ equal to $\tilde{\Sigma}_1$.

¹² This is also true when predicting returns and covariances of trading strategies, as shown in Table D1 in the appendix.

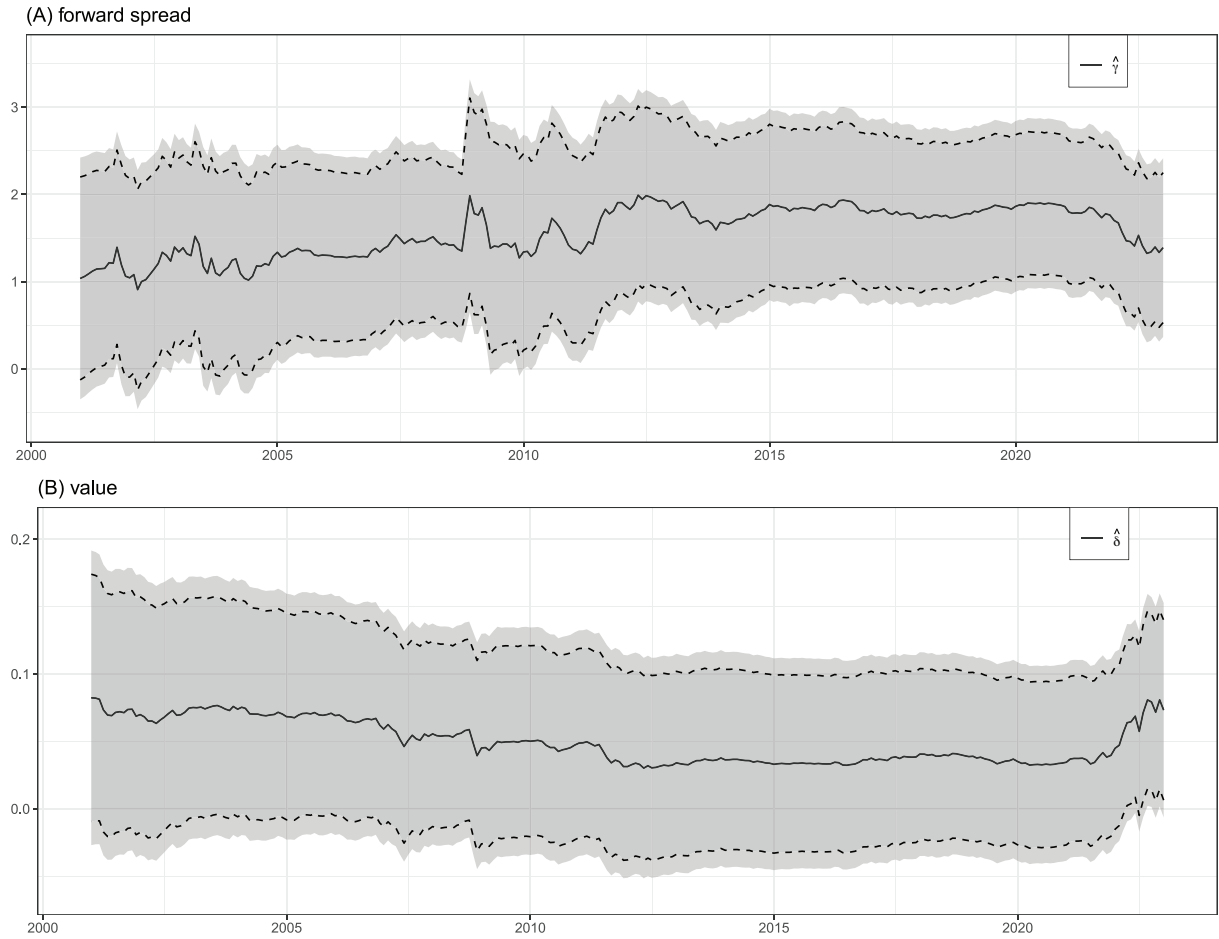


Fig. 2. Time series of estimated coefficients from predictive regression.

This figure shows the time series of the estimated coefficients from pooled regressions of the forward spread ($\hat{\gamma}$) and bond value ($\hat{\delta}$) on bond excess returns using expanding windows. Panel A displays the estimated coefficient of the forward spread variable and Panel B of the bond value variable. The dotted lines and the shaded areas show the 90% and 95% confidence intervals, respectively. Standard errors are clustered at the country level.

<i>Carry</i>	difference between the 10-year yield and the local 1-month yield $Carry_{i,t} = Y_{i,t}^{(10Y)} - Y_{i,t}^{(1M)}$,
<i>Value</i>	real bond yield, as defined in Eq. (9) $Value_{i,t} = Y_{i,t}^{(10Y)} - \pi_{i,t}$,
<i>Momentum</i>	cumulative return over 12 months, skipping the last month $Mom12_{i,t} = \prod_{k=1}^{11} \left(\frac{p_{i,t-k}^{(10Y-1M)}}{p_{i,t-k-1}^{(10Y)}} \frac{S_{i,t-k}}{S_{i,t-k-1}} \right) - 1$,
<i>ShortTermMom</i>	return over the last month $Mom1_{i,t} = \frac{p_{i,t}^{(10Y-1M)}}{p_{i,t-1}^{(10Y)}} \frac{S_{i,t}}{S_{i,t-1}} - 1$.

Table 5 lists the different trading strategies and provides the details of the respective portfolio constructions.

4.2. Regression tests of the UMVE portfolio

To analyze whether the UMVE portfolio prices international government bonds, we test Equation (2) and run traditional regression tests of the following form:

$$rx_{t+1}^i = \alpha_i + \beta_i rx_{t+1}^U + \epsilon_{i,t+1}, \quad (12)$$

where rx_{t+1}^i denotes the excess returns of test asset i (where i denotes the country or the trading strategy, respectively) and rx_{t+1}^U is the UMVE portfolio's excess return as defined in Section 2.

Table 3

Estimated coefficients from predictive regressions.

Statistic	Mean	Median	Std. Dev.	Pctl(10)	Pctl(90)
$\hat{\gamma}_t$	1.569	1.659	0.276	1.184	1.882
$t\text{-stat } \hat{\gamma}_t$	2.736	2.725	0.696	1.805	3.632
$\hat{\delta}_t$	0.050	0.045	0.016	0.033	0.072
$t\text{-stat } \hat{\delta}_t$	1.151	1.064	0.286	0.840	1.527

This table reports descriptive statistics (mean, median, standard deviation, 10th percentile, and 90th percentile) of the estimated coefficients from the pooled regressions specified in Eq. (7). $\hat{\gamma}_t$, $\hat{\delta}_t$ denote the estimated coefficients of the forward spread and value, respectively. $t\text{-stat}$ is the t -statistic of these variables.

Table 6 shows the results of the hypothesis test for the G10* markets. Specifically, it shows for each country the annualized Sharpe ratio, the mean excess returns, and the t -test of the mean excess returns. It further reports the alphas from country regressions defined in Eq. (12), their t -statistics, and the associated adjusted R^2 . While Sharpe ratios range from 0.26 to 0.68 and mean annualized excess returns between 2.17% and 4.05%, alphas are small and not significantly different from zero. The alpha for Norway has the largest t -statistic, with a value of 1.01. Japan exhibits the largest adjusted R^2 : the UMVE portfolio excess returns explain 17% of Japanese bond excess returns. Overall, results confirm the validity of the UMVE portfolio, as it is able to individually price government bond markets without bias. However, the relatively low R^2 s suggest that only a part of bond return variations is priced.

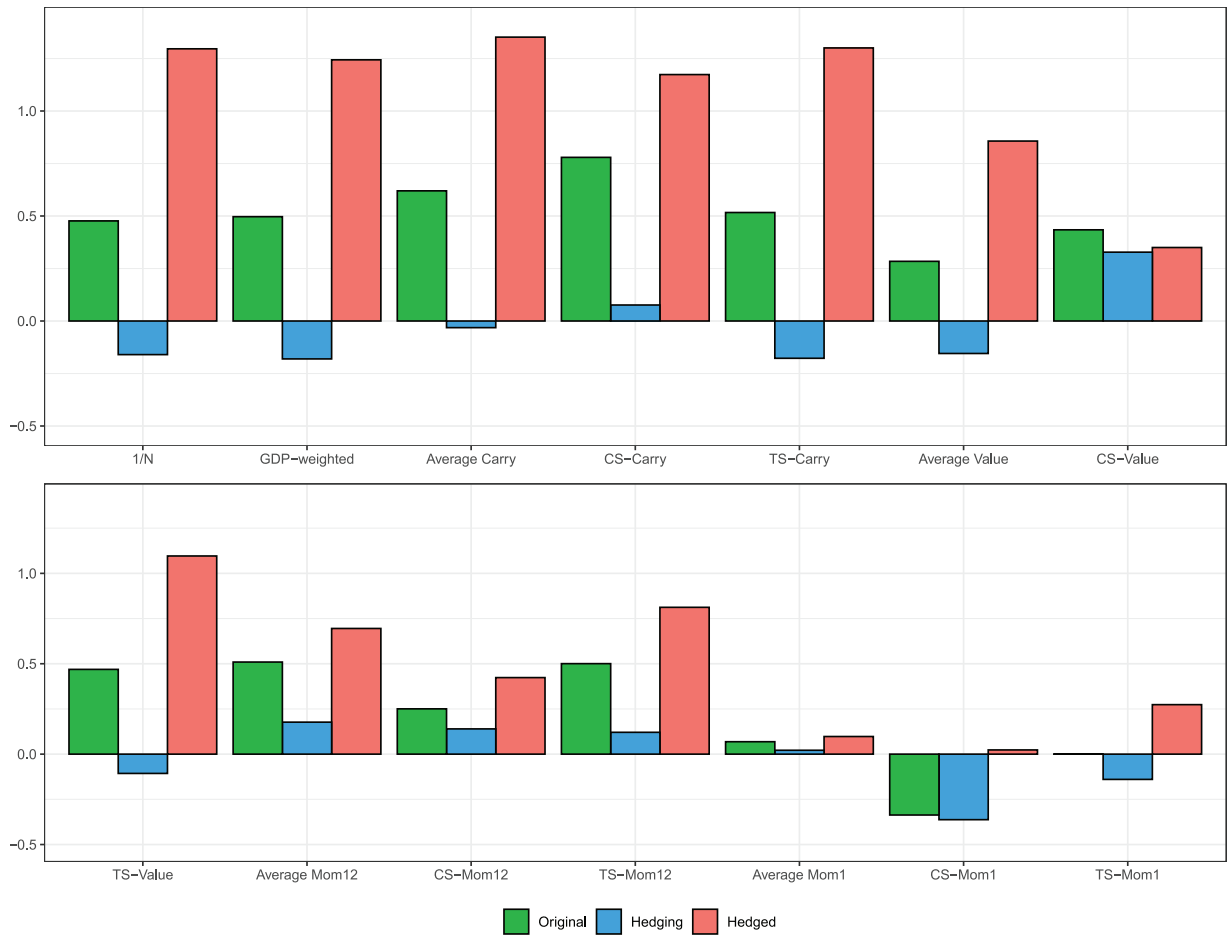


Fig. 3. Sharpe ratio of original, hedging, and hedged strategies.

This figure shows the annualized Sharpe ratio of the original, hedging and hedged portfolio strategies. The original portfolio strategies are defined in Table 5. The hedged portfolio is the portfolio with systematic factor exposure, as defined in Section 4.4, while the hedging portfolio captures the unpriced source of common variation in bond returns. Data are from January 2001 to December 2022.

Table 4
Predictive ability of conditional expectations, variance and covariance for individual countries.

	rx_{t+1}^i	$(rx_{t+1}^i - \mathbb{E}_t(rx_{t+1}^i))^2$	$(rx_{t+1}^i - \mathbb{E}_t(rx_{t+1}^i)) \cdot (rx_{t+1}^j - \mathbb{E}_t(rx_{t+1}^j))$
$\mathbb{E}_t(rx_{t+1}^i)$	0.911 (0.293)		
$\sigma_t^2(rx_{t+1}^i)$		0.950 (0.085)	
$cov_t(rx_{t+1}^i, rx_{t+1}^j)$			0.999 (0.115)
Observations	2,640	2,640	11,880
Adjusted R ²	0.011	0.057	0.054

This table shows estimates from regressions of realized returns, rx_{t+1}^i , on expected returns, $\mathbb{E}_t(rx_{t+1}^i)$, column (1), of $(rx_{t+1}^i - \mathbb{E}_t(rx_{t+1}^i))^2$ on estimated variance, $\sigma_t^2(rx_{t+1}^i)$, column (2), and of $(rx_{t+1}^i - \mathbb{E}_t(rx_{t+1}^i)) \cdot (rx_{t+1}^j - \mathbb{E}_t(rx_{t+1}^j))$ on estimated covariance, column (3). The estimation of the conditional means and the conditional covariances follows Sections 3.4.1 and 3.4.2, respectively. Standard errors are clustered at the month level. Under the null hypothesis that coefficients are equal to 1, ***, **, * indicates that estimates are statistically significant at the 1, 5, and 10 percent levels, respectively.

This observation is consistent with the finding of Chernov et al. (2023) for currencies.

Table 7 reports the results for portfolio strategies. The Sharpe ratios of the 14 portfolio strategies range from -0.34 to 0.78 . All alphas with respect to the UMVE portfolio are statistically insignificant. The UMVE portfolio prices a relatively large share of the variation in the cross-sectional carry strategy, as expressed by the R^2 of 21%. The other R^2 s are relatively low, similar in magnitude to those of Table 6. This suggests again that a substantial fraction of the variation of the returns is unpriced.

Table 8 reports the results of the Gibbons-Ross-Shanken (GRS) test (Gibbons et al., 1989) and the multihorizon returns (MHR) test (Chernov et al., 2022). The first column shows the Sharpe ratio of the UMVE portfolio, constructed from the full model defined in Eq. (7) (top row), and of alternative UMVE portfolios, which, alternatively, only use the forward spread (middle row) or value (bottom row) as predictor variable. As can be seen, the UMVE portfolio based on the full model prices bond markets substantially better than the UMVE portfolio obtained from the restricted models. This indicates that, while the bond value variable exhibits poor statistical significance when we look at

Table 5
Trading strategies.

Acronym	Name	Description
1/ <i>N</i>	Naive	equally-weighted average of countries' bond returns
GDP-weighted	GDP-weighted	GDP-weighted average of countries' bond returns
Average Carry	Average Carry	long (short) in all countries when the average bond carry is positive (negative)
CS-Carry	Cross-Sectional Carry	rank weights (Asness et al., 2013) based on bond carry
TS-Carry	Time Series Carry	long (short) in countries with a positive (negative) bond carry
Average Value	Average Value	long (short) in all countries when the average bond value is positive (negative)
CS-Value	Cross-Sectional Value	rank weights based on bond value
TS-Value	Time Series Value	long (short) in countries with a positive (negative) bond value
Average Mom12	Average Momentum 12 Months	long (short) in all countries when the average long-term bond momentum is positive (negative)
CS-Mom12	Cross-Sectional Momentum 12 Months	rank weights based on long-term bond momentum
TS-Mom12	Time Series Momentum 12 Months	long (short) in countries with a positive (negative) long-term bond momentum
Average Mom1	Average Momentum 1 Month	long (short) in all countries when the average short-term bond momentum is positive (negative)
CS-Mom1	Cross-Sectional Momentum 1 Month	rank weights based on short-term bond momentum
TS-Mom1	Time Series Momentum 1 Month	long (short) in countries with a positive (negative) short-term bond momentum

Table 6
Testing the UMVE using individual country bond returns.

	SR	Mean	t_mean	Alpha	t_alpha	adj.R ²
Australia	0.259	2.176	1.219	0.871	0.461	0.012
Canada	0.466	3.237	2.190	0.698	0.454	0.085
Germany	0.500	3.471	2.348	0.985	0.765	0.081
Japan	0.684	2.272	3.212	0.569	0.808	0.171
New Zealand	0.290	2.341	1.364	0.433	0.244	0.033
Norway	0.384	2.754	1.803	1.578	1.014	0.014
Sweden	0.456	3.287	2.142	1.156	0.792	0.054
Switzerland	0.508	2.911	2.386	0.839	0.709	0.083
United Kingdom	0.313	2.539	1.472	0.372	0.223	0.044
United States	0.443	4.046	2.083	0.733	0.371	0.084

This table shows, for individual countries, the annualized Sharpe ratio (*SR*), mean bond returns (*Mean*, in % p.a.), and the *t*-statistic for the mean bond returns (*t_mean*). It further reports the alpha from regressing individual countries' bond returns on the UMVE portfolio returns (*Alpha*, in % p.a.), the *t*-statistics for the alpha coefficient (*t_alpha*), and the adjusted R² from this regression (*adj.R²*). The *t*-statistics are heteroskedasticity-robust. Data are from January 2001 to December 2022.

Table 7
Testing the UMVE using portfolio strategies.

	SR	Mean	t_mean	Alpha	t_alpha	adj.R ²
1/ <i>N</i>	0.477	2.903	2.239	0.823	0.660	0.074
GDP-weighted	0.497	3.350	2.334	0.726	0.522	0.097
Average Carry	0.620	3.752	2.913	1.272	1.024	0.108
CS-Carry	0.779	3.016	3.662	0.817	1.040	0.211
TS-Carry	0.517	2.782	2.428	0.729	0.671	0.093
Average Value	0.284	1.742	1.336	0.299	0.236	0.033
CS-Value	0.434	1.422	2.041	0.869	1.256	0.015
TS-Value	0.469	2.287	2.203	0.775	0.758	0.060
Average Mom12	0.509	3.100	2.394	1.221	0.947	0.060
CS-Mom12	0.250	0.829	1.177	0.828	1.050	−0.004
TS-Mom12	0.501	2.531	2.352	1.240	1.149	0.040
Average Mom1	0.069	0.422	0.323	1.211	0.939	0.007
CS-Mom1	−0.337	−1.158	−1.583	−1.458	−1.851	0.001
TS-Mom1	0.001	0.005	0.005	0.642	0.643	0.008

This table shows for global bond portfolio strategies, as defined in Table 5, the annualized Sharpe ratio (*SR*), mean bond returns (*Mean*, in % p.a.), and the *t*-statistic for the mean bond returns (*t_mean*). It further shows the alpha from regressing portfolio returns on the UMVE portfolio returns (*Alpha*, in % p.a.), the *t*-statistics for the alpha coefficient (*t_alpha*), and the adjusted R² from this regression (*adj.R²*). The *t*-statistics are heteroskedasticity-robust. Data are from January 2001 to December 2022.

its effect in the forecasting regression (see Table 3), it is an important variable, jointly with the forward spread, when it comes to pricing the cross-section of bond returns. We also observe that the Sharpe ratio of the optimal UMVE portfolio is much higher than the Sharpe ratios of the individual bond markets and those of bond strategies.

Columns 2 and 3 of Table 8 show the results from asset pricing tests using all strategies as test assets. We report the *p*-values of the tests whether the alphas from regressing each portfolio strategy on the

UMVE portfolio are jointly equal to zero. Both the GRS and MHR tests fail to reject the null hypothesis with *p*-values equal to 0.455 and 0.106, respectively. If we consider simpler models using the forward spread only (middle row), or value only (bottom row), the GRS test rejects the null with a *p*-value equal to 0.047 for the model using the forward spread and 0.012 for the model using value. Similarly, the MHR test rejects the null hypothesis for the restricted models. When using carry and value strategies as test assets (columns 4 and 5), the GRS and MHR

Table 8
Sharpe ratios and asset pricing tests using portfolio strategies.

Model	UMVE	All strategies		Carry and value	
	SR	GRS	MHR	GRS	MHR
Optimal	1.229	0.455	0.106	0.544	0.718
Forward spread only	0.969	0.047	0.010	0.044	0.688
Value only	0.688	0.012	0.019	0.003	0.036

This table shows in column (1) the Sharpe ratio (SR) of the UMVE portfolio constructed using the full estimation model, i.e. based on the forward spread and value as predictor variables (top row), and of restricted UMVE portfolios constructed using only the forward spread (middle row) or using only value (bottom row) as predictor variable. Columns (2) and (3) show p-values of the GRS-test and MHR test, respectively, using all strategies as test assets. Columns (4) and (5) show p-values of the GRS-test and MHR test, respectively, using carry and value strategies as test assets. Data are from January 2001 until December 2022.

tests again do not reject the optimal model, while the evidence does not support the restricted models.

Overall, the UMVE portfolio, constructed using both the forward spread and value signals, has the highest Sharpe ratio and is the only model not rejected in any of the tests. Therefore, we conclude that the full estimation model represents a valid SDF.

4.3. Pricing other countries

Since a global SDF should not only price the assets from which it has been constructed, we go beyond the original set of the G10* markets, and implement tests using a wider universe of bond markets. Specifically, we test if the UMVE portfolio prices single-country bonds and strategies based on an extended set of countries, which are not used to construct the UMVE portfolio. We therefore run regression tests, as specified in Section 4.2, using the excess returns of other countries' bond markets and the trading strategies associated with most of these markets¹³ as the dependent variable in Eq. (12).

We report the results in Table 9. While, on average, alphas are larger in magnitude compared to the results for the main markets (see Table 6), none is statistically significant at the 5% level. This is confirmed both for markets pegged to the Euro and, importantly, also for other markets. The regression for Italy exhibits the largest *t*-stat, with a value of 1.65. This might reflect the different behavior of Italian bonds during and after the European sovereign debt crisis, when their status as a safe asset was in doubt. The picture does not materially change when we consider bond strategies constructed from this set of bonds. Table 10 shows that most alphas are insignificant (at the 5% level), with the exception of cross-sectional carry and cross-sectional 12-month momentum. In summary, there is little evidence of market segmentation, as the UMVE portfolio constructed from the main bond markets does not generally lead to large pricing errors even in a broader asset universe.

4.4. Unpriced return components

Our results so far confirm that the UMVE portfolio prices both individual countries' bond excess returns and the excess returns obtained from portfolio strategies. However, the share of variation in returns explained by the UMVE, as expressed by the regressions R^2 s, is relatively low. We therefore construct bond portfolios that hedge out the unpriced sources of common variation in returns. The recent literature explores the properties of hedged and unhedged portfolios

and their asset pricing implications (e.g., Daniel et al., 2020; Lopez-Lira and Roussanov, 2023). These papers find that Sharpe ratios improve substantially when hedging unpriced components of risk. We compute for each strategy the conditional beta, β_t^P , with respect to the UMVE portfolio as follows:

$$\beta_t^P = \frac{(\mathbf{w}_t^P)^\top \nabla_t(\mathbf{r}\mathbf{x}_{t+1})\mathbf{w}_t^U}{(\mathbf{w}_t^U)^\top \nabla_t(\mathbf{r}\mathbf{x}_{t+1})\mathbf{w}_t^U}, \quad (13)$$

where $\mathbf{w}_t^P$ is a vector of strategy weights, as defined in Table 5, and $\mathbf{w}_t^U$ is the vector with the weights of the UMVE portfolio.

We define the hedged portfolio, HDP , as a portfolio with excess returns given by $\beta_t^P r x_{t+1}^U$, i.e. as a portfolio which only has systematic factor exposure. The unpriced component is captured by the hedging portfolio, HGP , with returns equal to the return difference between the original strategy portfolio, P , and the hedged portfolio, HDP . In particular:

$$r x_{t+1}^{HGP} = r x_{t+1}^P - \beta_t^P r x_{t+1}^U. \quad (14)$$

The weights of the hedged portfolios are given by $\mathbf{w}_t^{HDP} = \beta_t^P \cdot \mathbf{w}_t^U$. In Appendix E, we provide figures plotting the time series of portfolio weights of the respective hedged trading strategy.

Fig. 3 displays the annualized Sharpe ratios of the original portfolios, of the hedging portfolios and of the hedged portfolios. The Sharpe ratios of most hedging portfolios are close to zero, and always lower than those of the original strategies.¹⁴ This accords well with the notion that risks, which are not related to the SDF, should not be associated with risk premia. We also observe that for most strategies the Sharpe ratios of the hedged portfolios are all substantially larger than those of the hedging and the original portfolio strategies, respectively.¹⁵ This is consistent with Daniel et al. (2020), who conclude that the risk-adjusted portfolio return can be improved by eliminating unpriced risks. Indeed, while the risk-adjusted portfolio returns for the original strategies are in the range of -0.34 to 0.78 , those of the hedged portfolios range between 0.02 and 1.35 .

To gain additional insight into the structure of bond returns and their relation to the UMVE portfolio, we perform a Principal Components Analysis (PCA) of bond excess returns. Fig. 4 displays the loadings of the first three principal components (PCs). The first PC is a level factor, as all countries load positively on it. The second PC seems to be an Australasia vs. rest of the world factor. Australia and New Zealand have indeed positive loadings, while the remaining countries have negative ones. With respect to the third PC, we cannot see a clear, intuitive interpretation.

Fig. 5 documents how much of the variation in the UMVE portfolio returns is explained by the PCs. Panel B of Fig. 5 reports the R^2 of these regressions. The first bar shows the results from regressing the UMVE on the first PC only, the second one uses the first two PCs as regressors and similar interpretations apply for the other bars. Interestingly, the first three PCs explain less than 10% of the variation in the UMVE excess returns. This is despite the fact that these three PCs jointly explain around 86% of the variation in bond returns, as documented in Panel A of Fig. 5. Even when all PCs are considered, the R^2 only increases to 28%. This implies that the UMVE is not spanned by the PCs, since a significant fraction of its variation remains unexplained. Panel C of Fig. 5 shows the alphas from a regression of bond returns on the PCs, as well as their confidence intervals. It can be seen that

¹³ To guarantee a balanced panel and data availability from January 2001 (starting date of the UMVE) for the trading strategies, we use only data on the following markets to construct the alternative countries' portfolio strategies: Austria, Belgium, Denmark, Finland, France, Hong Kong, Ireland, Italy, Netherlands, Portugal, Singapore, and Spain.

¹⁴ Testing for statistical significance using Lo (2002), we do not reject the null hypothesis of a Sharpe ratio equal to zero for 13 out of the 14 hedging portfolios. Only CS-Mom1 is marginally statistically significant with a *p*-value of 0.09.

¹⁵ For short-term cross-sectional momentum, the absolute magnitude Sharpe ratio of the hedging portfolio exceeds the original portfolio's Sharpe ratio. However, both values are negative, so even in this case hedging leads to a higher Sharpe ratio.

Table 9
Testing the UMVE using alternative countries' bond returns.

	SR	Mean	t_mean	Alpha	t_alpha	adj.R ²
<i>Euro area</i>						
Austria	0.522	3.754	2.453	1.272	0.952	0.075
Belgium	0.536	4.154	2.517	1.520	1.051	0.073
Finland	0.493	3.535	2.317	0.996	0.741	0.080
France	0.546	3.940	2.567	1.647	1.213	0.063
Greece	0.274	10.469	1.286	11.439	1.394	−0.003
Ireland	0.463	5.130	2.174	2.809	1.113	0.025
Italy	0.531	5.153	2.496	3.330	1.652	0.020
Netherlands	0.511	3.661	2.400	1.170	0.880	0.076
Portugal	0.434	5.740	2.040	3.920	1.368	0.009
Slovakia*	0.431	3.516	1.769	2.062	0.966	0.019
Slovenia*	0.322	3.267	1.372	2.234	0.873	0.004
Spain	0.546	4.890	2.563	2.652	1.393	0.038
<i>Other countries</i>						
Chile*	0.286	2.989	1.188	2.381	0.922	−0.002
Denmark	0.478	3.497	2.247	1.087	0.777	0.068
Colombia*	0.251	3.628	1.030	−0.410	−0.115	0.055
Czech Republic	0.405	3.706	1.901	1.519	0.770	0.034
Hong Kong	0.437	4.309	2.052	1.810	0.820	0.039
Hungary*	0.231	3.536	1.080	1.006	0.300	0.014
Israel*	0.577	5.262	2.444	2.482	1.133	0.074
Mexico*	0.305	3.822	1.348	0.781	0.278	0.037
Poland	0.305	3.502	1.434	1.883	0.752	0.009
Singapore	0.439	3.271	2.062	0.675	0.413	0.077
Turkey*	0.348	12.138	1.468	3.711	0.450	0.046

This table shows for other countries, which we do not use for the estimation of the UMVE portfolio, the annualized Sharpe ratio (*SR*), mean bond returns (*Mean*, in % p.a.), and the *t*-statistic for the mean bond returns (*t_mean*). It further shows the alpha from regressing these countries' bond returns on the UMVE portfolio returns (*Alpha*, in % p.a.), the *t*-statistics for the alpha (*t_alpha*), and the adjusted *R*² from this regression (*adj.R*²). Data are from January 2001 until December 2022. * indicates countries for which data are available for a shorter time period. Specifically, data availability starts as follows: Chile: September 2005, Colombia: April 2006, Hungary: March 2001, Israel: March 2005, Mexico: August 2003, Slovakia: April 2006, Slovenia: December 2004, Turkey: April 2005.

Table 10
Testing the UMVE using alternative countries' portfolio strategies.

	SR	Mean	t_mean	Alpha	t_alpha	adj.R ²
1/ <i>N</i>	0.616	4.253	2.894	1.907	1.419	0.073
GDP-weighted	0.609	4.323	2.860	2.067	1.490	0.063
Average Carry	0.616	4.253	2.894	1.907	1.419	0.073
CS-Carry	0.642	4.634	3.016	4.647	2.616	−0.004
TS-Carry	0.609	4.041	2.863	1.922	1.454	0.064
Average Value	0.681	4.688	3.201	1.594	1.061	0.131
CS-Value	0.436	2.992	2.051	2.881	1.754	−0.004
TS-Value	0.756	4.154	3.555	2.041	1.682	0.095
Average Mom12	0.736	5.052	3.460	2.870	1.985	0.063
CS-Mom12	0.535	3.324	2.516	4.248	2.866	0.011
TS-Mom12	0.692	4.138	3.250	2.443	1.965	0.050
Average Mom1	0.320	2.232	1.502	2.787	1.859	0.000
CS-Mom1	−0.074	−0.410	−0.346	0.305	0.239	0.007
TS-Mom1	0.193	1.154	0.905	1.723	1.418	0.002

This table shows for global bond portfolio strategies, as defined in Table 5, of other countries, which we do not use for the estimation of the UMVE portfolio, the annualized Sharpe ratio (*SR*), mean bond returns (*Mean*, in % p.a.), and the *t*-statistic for the mean bond returns (*t_mean*). It further shows the alpha from regressing portfolio strategies of these countries' bond returns on the UMVE portfolio returns (*Alpha*, in % p.a.), the *t*-statistics for the alpha coefficient (*t_alpha*), and the adjusted *R*² from this regression (*adj.R*²). Data are from January 2001 until December 2022.

the resulting alphas are all positive and statistically significant. Thus, PCs extracted from bond returns do not adequately explain the UMVE portfolio returns.

Overall, this evidence suggests that there is an important timing component in the UMVE portfolio, which is not reflected in the PCs of bond returns. The latter are, indeed, essentially a linear combination of individual bond returns weighted by constant loadings, and as such cannot capture the dynamic nature of the UMVE portfolio. Such a timing component is likely to be even more strongly reflected in trading strategies. We therefore turn our attention to dynamic trading strategies next.

Fig. 6 shows the PC loadings of the (dynamic) bond trading strategies. These loadings do not imply a straightforward economic interpretation of the PCs, except for the first PC, which again seems to reflect a level factor. As displayed in Panel A of Fig. 7, the first three PCs explain 78% of the variation in portfolio strategies. However, even though these three PCs capture the time variation of strategy returns quite well, they only explain a small share of priced risks, i.e. 14%, as documented in Panel B of Fig. 7. If we use all PCs, the *R*² increases to 33%. As is the case for individual bonds, we document that the factors explaining the variation in the returns on trading strategies are not important drivers of the variation in the UMVE portfolio returns. However, consistent with the fact that trading strategies better capture

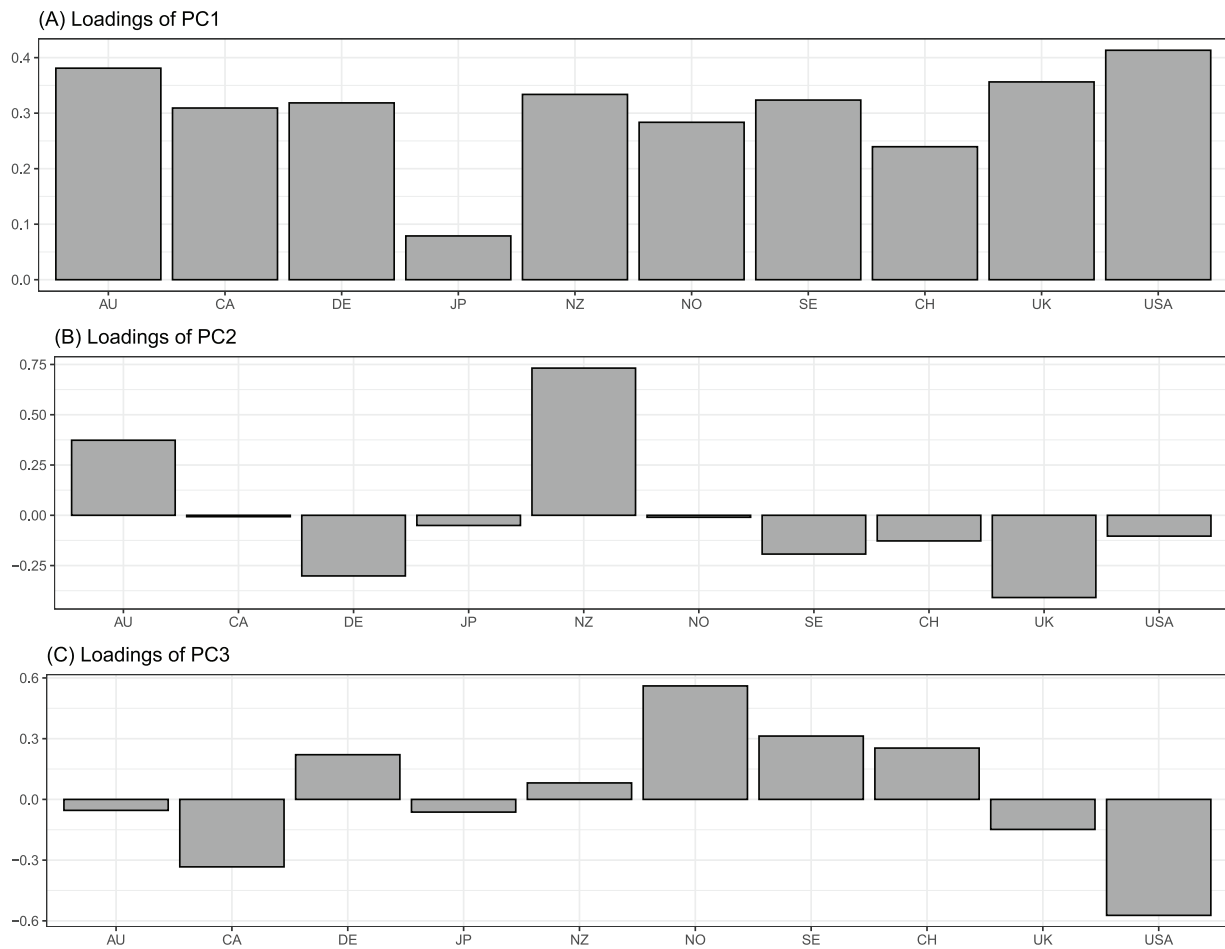


Fig. 4. Principal component analysis of returns: Loadings.

This figure shows loadings from Principal Component Analysis of bond returns. Bond returns refer to the G10* countries. Panel A reports loadings for PC1, Panel B for PC2, and Panel C for PC3. Data are from January 2001 to December 2022.

Table 11

Descriptive statistics for the UMVE portfolio.

	Full sample period
Mean (in %)	7.485
Median (in %)	7.047
Std. Dev. (in %)	6.092
Skewness	0.527
Kurtosis	4.447
Sharpe Ratio	1.229
Maximum Drawdown (in %)	9.917
$w_{i,t}^{U,+}$ (in %)	41.914
$w_{i,t}^{U,-}$ (in %)	-34.293
$ w_{i,t}^U $ (in %)	38.502
$\sum w_{i,t}^U - w_{i,t-1}^U $	1.543

This table shows descriptive statistics for the UMVE portfolio. UMVE returns are scaled so that they have the same volatility as the naive portfolio. Specifically, it shows the mean excess return (in % p.a.), median excess return (in % p.a.), standard deviation (in % p.a.), Sharpe ratio (annualized), and maximum drawdown (in %). The second set of rows provides statistics for the UMVE portfolio weights averaged across time and markets. It reports the average absolute portfolio weight (in %), the average of positive weights (in %), the average of negative weights (in %), and the average portfolio turnover. Data are from January 2001 to December 2022.

Table 12

Descriptive statistics of the UMVE portfolio for Pre- and Post-08/2008 periods.

	Pre-08/2008	Post-08/2008
Mean (in %)	7.421	7.519
Median (in %)	7.086	6.937
Std. Dev. (in %)	4.783	6.692
Skewness	0.805	0.454
Kurtosis	5.687	7.064
Sharpe Ratio	1.552	1.124
Maximum Drawdown (in %)	5.202	9.917
$w_{i,t}^{U,+}$ (in %)	35.767	45.231
$w_{i,t}^{U,-}$ (in %)	-29.496	-36.738
$ w_{i,t}^U $ (in %)	33.017	41.387
$\sum w_{i,t}^U - w_{i,t-1}^U $	1.191	1.726

This table shows descriptive statistics for the UMVE portfolio. UMVE returns are scaled so that they have the same volatility as the naive portfolio. Specifically, it shows the mean excess return (in % p.a.), median excess return (in % p.a.), standard deviation (in % p.a.), Sharpe ratio (annualized), and maximum drawdown (in %). The second set of rows provides statistics for the UMVE portfolio weights averaged across time and markets. It reports the average absolute portfolio weight (in %), the average of positive weights (in %), the average of negative weights (in %), and the average portfolio turnover. Data for the columns labeled *Pre-08/2008* cover the period from January 2001 to July 2008. *Post-08/2008* covers the period from August 2008 to December 2022.

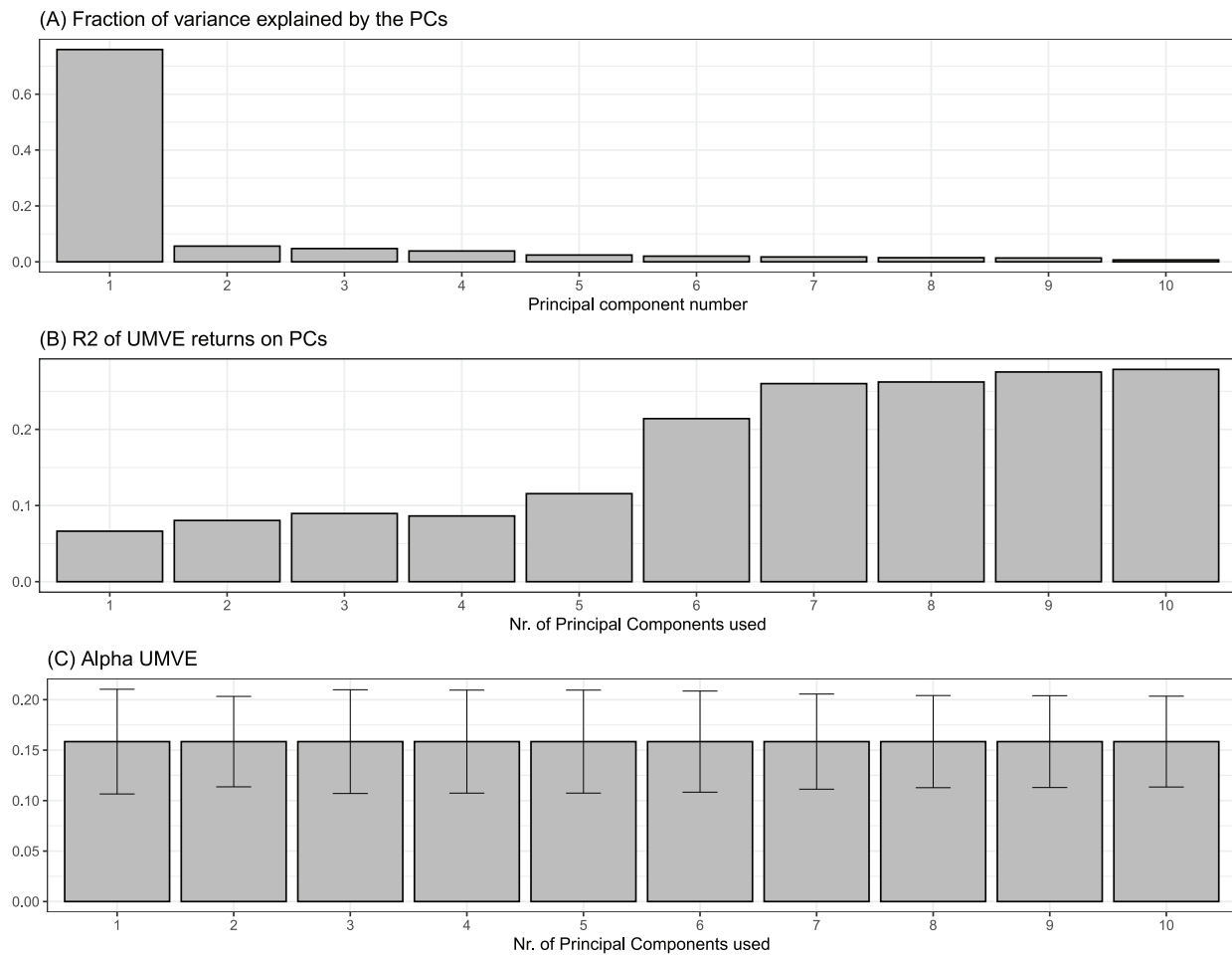


Fig. 5. Principal component analysis of bond returns and UMVE returns.

This figure shows in Panel A the fraction of variance explained by the PCs of bond returns. Bond returns refer to the G10* countries. Panel B shows the adjusted R^2 from regressing the UMVE returns on the PCs of bond returns. Panel C shows the alpha from these regressions, the black lines are the 95% confidence intervals. Data are from January 2001 to December 2022.

the dynamic nature of the UMVE, the principal components obtained from these portfolios explain a higher fraction of the UMVE portfolio returns than the ones obtained from the set of individual bond markets.

4.5. Sharpe ratio decomposition

Section 4.4 documents that the Sharpe ratios of almost all traditional trading strategies can be substantially improved by hedging out risk components unrelated to UMVE portfolio returns. This section analyzes whether this improvement is mainly due to the SDF capturing the cross-sectional and time series variation of the various markets' expected returns, the time-varying return dependence, or to both being of similar importance.

To this end, we successively shut off elements in the estimation of the UMVE portfolio (i.e., expected returns, variances, correlations) and only let the others vary. Comparing hedging strategies based on the restricted UMVE portfolios helps to identify the sources of the Sharpe ratio improvements. Specifically, our approach considers the following alternatives: First, we fix expected returns at the historical average across all markets over the estimation window, but use the estimates for the conditional variance from the full model (as described in Section 3.4.2). Thus, we shut off any cross-sectional variation in conditional expected returns when constructing the UMVE portfolio. Second, we use estimates for expected returns for each market from the full model (as described in Section 3.4.1), but employ a covariance matrix $\mathbb{V}_t^{\sigma}$ where we set each market's variance equal to the cross-sectional average. Thus, here we shut off any cross-sectional variation

in conditional variances when constructing the UMVE portfolio. Third, we again estimate expected returns based on the full model (as described in Section 3.4.1), but use a covariance matrix $\mathbb{V}_t^{\rho}$ where we set all correlations equal to the average estimated pairwise correlation. Thus, here we shut off any cross-sectional variation in conditional correlations. Note that all three described estimates are out-of-sample since the values at which a component of the estimation is fixed only depends on past returns. We describe our approach in more detail in Appendix F.

We refer to the hedged portfolios resulting from the three restricted SDF estimations as *Hedged: Constant Expected Returns*, *Hedged: Constant Sigma* and *Hedged: Constant Correlation*. Fig. 8 shows the annualized Sharpe ratios for these portfolios. For easier interpretation, the first bar displays the Sharpe ratios of the hedged portfolios based on the unrestricted UMVE portfolio returns, as already shown in Fig. 3. As can be seen from Fig. 8, only the portfolios *Hedged: Constant Correlation* have Sharpe ratios close to the ones obtained from the hedged strategies that make full use of the estimated parameters, as is done in Fig. 3. In contrast, Sharpe ratios are much lower when using an SDF that is estimated from the UMVE portfolios that assume constant expected returns or constant standard deviations across markets. When comparing the second (green) and third (turquoise) bars in the charts for the different strategies, we see that modeling the cross-section of expected returns is more important in defining the UMVE portfolio than estimating the cross-section of variances. Indeed, the hedged portfolios when using UMVE portfolio estimations based on constant variances have higher

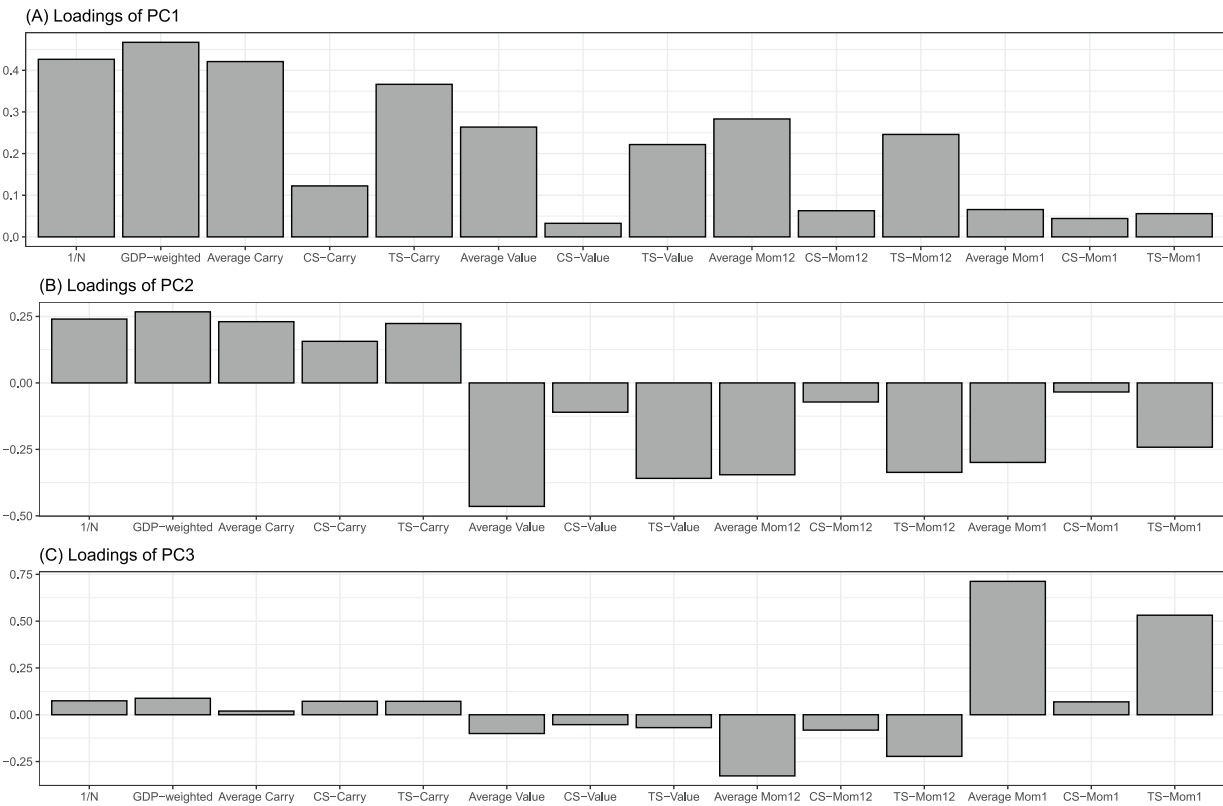


Fig. 6. Principal component analysis of strategies: Loadings. This figure shows loadings from Principal Component Analysis of global bond strategy returns. The portfolio strategies are defined in Table 5. Panel A reports loadings for PC1, Panel B for PC2, and Panel C for PC3. Data are from January 2001 to December 2022.

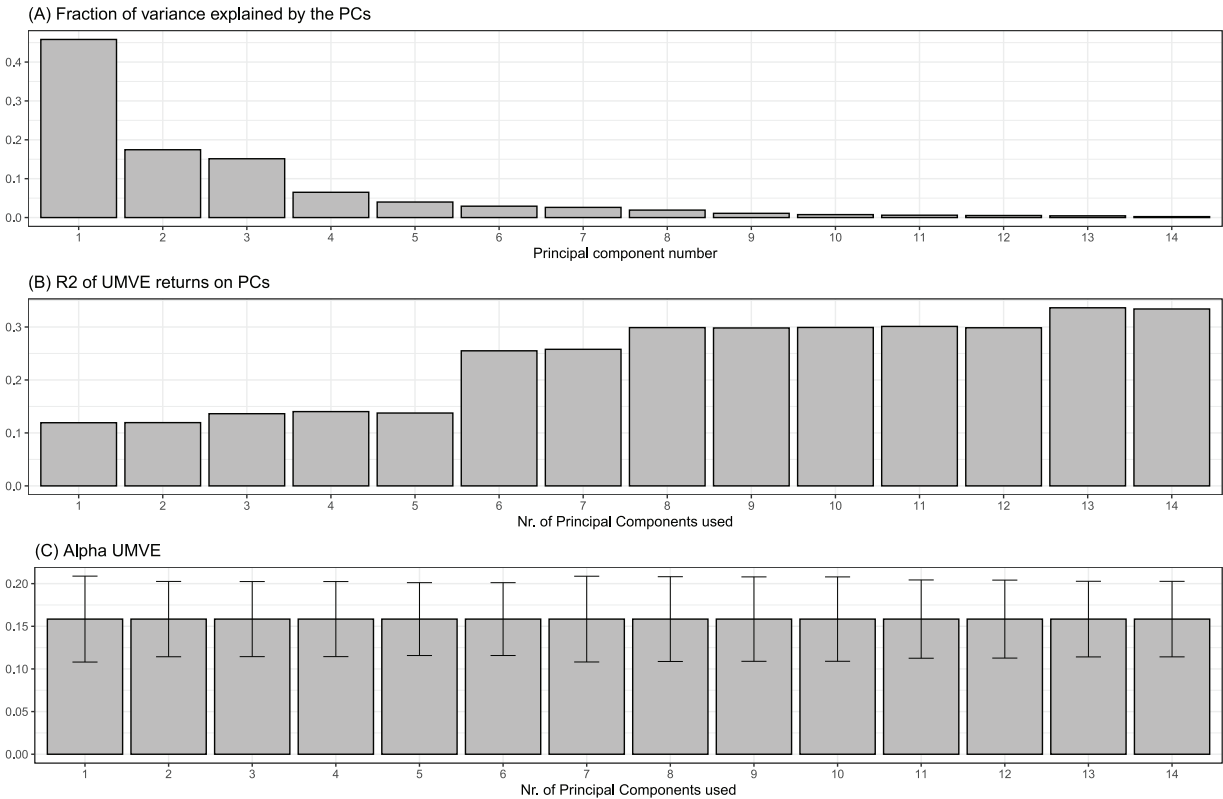


Fig. 7. Principal component analysis of strategies and UMVE returns. This figure shows in Panel A the fraction of variance explained by the PCs of global bond strategy returns. The portfolio strategies are defined in Table 5. Panel B shows the adjusted R² from regressing the UMVE returns on the PCs. Panel C shows the alpha from these regressions, the black lines are the 95% confidence intervals. Data are from January 2001 to December 2022.

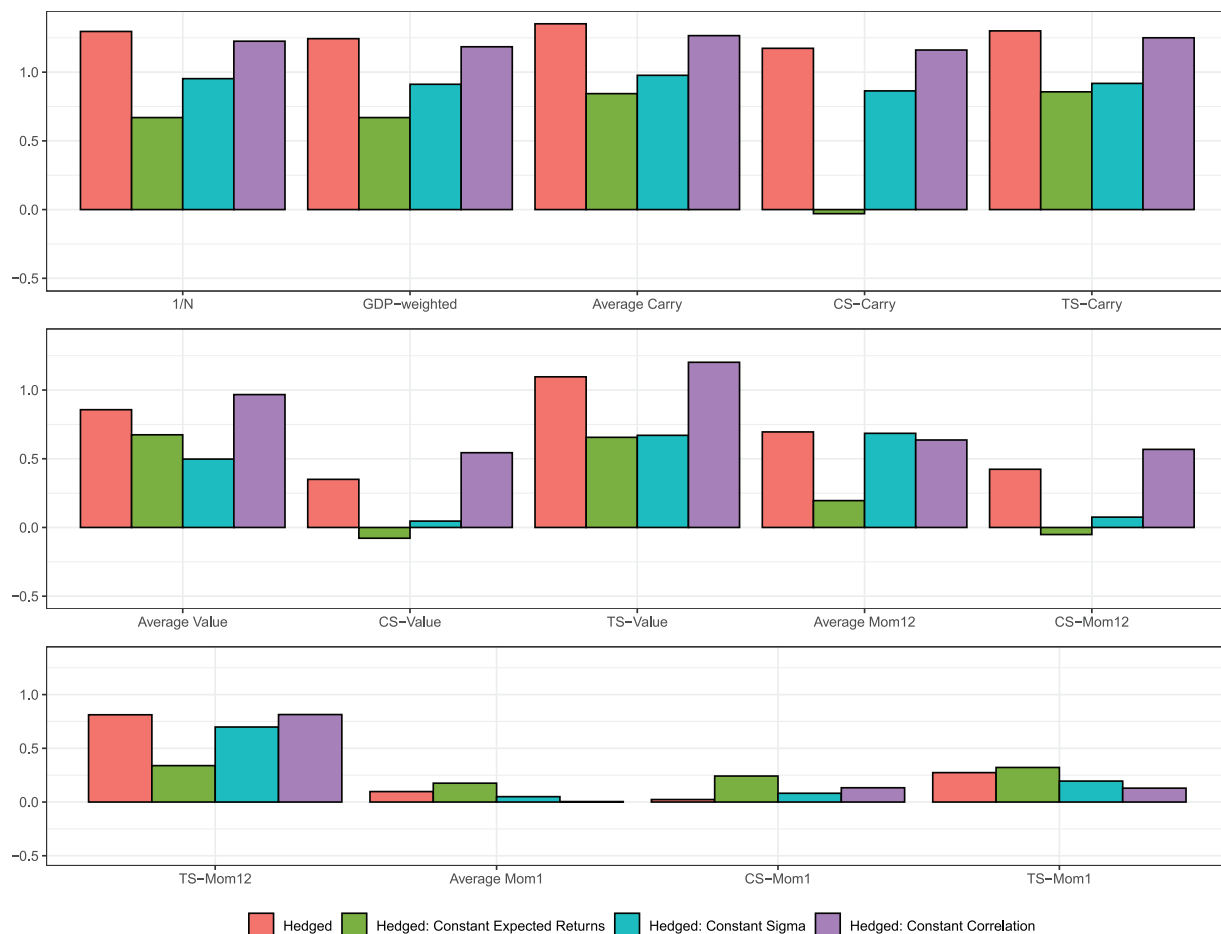


Fig. 8. Sharpe ratio hedged strategies decomposition.

This figure shows, for each strategy, the annualized Sharpe ratio of the hedged portfolio (first bar), as defined in Section 4.4, along with three alternative hedged portfolios based on restricted versions of the UMVE portfolio. We construct the restricted versions by successively holding constant one component – expected returns, variances, or correlations – while estimating the remaining components from the full model (as described in Sections 3.4.1 and 3.4.2). Specifically, the restricted portfolios eliminate cross-sectional variation in conditional expected returns, conditional variances, or conditional correlations, as described in Section 4.5. The resulting hedged portfolios are labeled *Hedged: Constant Expected Returns*, *Hedged: Constant Sigma*, and *Hedged: Constant Correlation*. Data are from January 2001 to December 2022.

Sharpe ratios on average than in the case where UMVE portfolios are constructed based on constant expected returns, indicating that shutting off cross-sectional variation in expected returns worsens the performance of the resulting SDF more than shutting off cross-sectional variation in variances.¹⁶

4.6. UMVE portfolio characteristics

This section reports important properties of the UMVE portfolio shown to price bond markets. In particular, we provide descriptive statistics and analyze its time-varying risk-return characteristics. To this

¹⁶ The analysis in this subsection eliminates various sources of cross-sectional differences in the parameters that determine the UMVE portfolio. For completeness, Appendix G reports results of an alternative exercise, which eliminates various sources of time series differences instead. Specifically, we (i) use the in-sample country-specific average excess returns, but let the conditional variance be estimated as in Section 3.4.2 and (ii) we let expected returns be estimated as in Section 3.4.1, but use the in-sample estimated covariance matrix, by computing the arithmetic average of all monthly shrunk covariance matrices. The detailed description of this approach as well as the results are available in Appendix G. Overall, the outcome is consistent with the conclusions of this subsection in the sense that capturing time series differences in expected returns is more important than capturing time series differences in covariances.

purpose, we scale the UMVE portfolio, so that its returns have the same average volatility as the 1/N portfolio, i.e. a portfolio that allocates capital equally across all bond markets.

Table 11 summarizes descriptive statistics. Specifically, it shows that the UMVE portfolio returns an impressive average annualized excess return of 7.5% with a volatility of 6.1% over the entire period from January 2001 to December 2022. This implies a Sharpe ratio of 1.23. The maximum drawdown is 9.9%. Returns are positively skewed (skewness equals 0.53) and exhibit fat tails (kurtosis is equal to 4.4). The second set of rows shows average weights (across markets and time) of the optimal portfolio weights. It can be seen that individual positions are quite large: the average absolute weight is about four times that of the 1/N portfolio. The dynamics of the portfolio composition are illustrated in Fig. 9, which plots the weights of the constituents of the UMVE portfolio over time. There is considerable time-variation in the portfolio composition, which is consistent with Green and Hollifield (1992), who argue that the presence of a dominant factor can lead to extreme weights of assets in mean-variance efficient portfolios. The sensitivity of results with respect to restrictions on portfolio turnover is discussed in Section 6.

Fig. 10 plots the UMVE's annualized expected Sharpe ratio over time. The figure illustrates that the market price of risk exhibits significant time-variation. The highest expected Sharpe ratio is 2.68 and it reaches a minimum of 0.74. We observe relatively high Sharpe ratios in periods of crises, as reflected by the peaks around the financial crisis

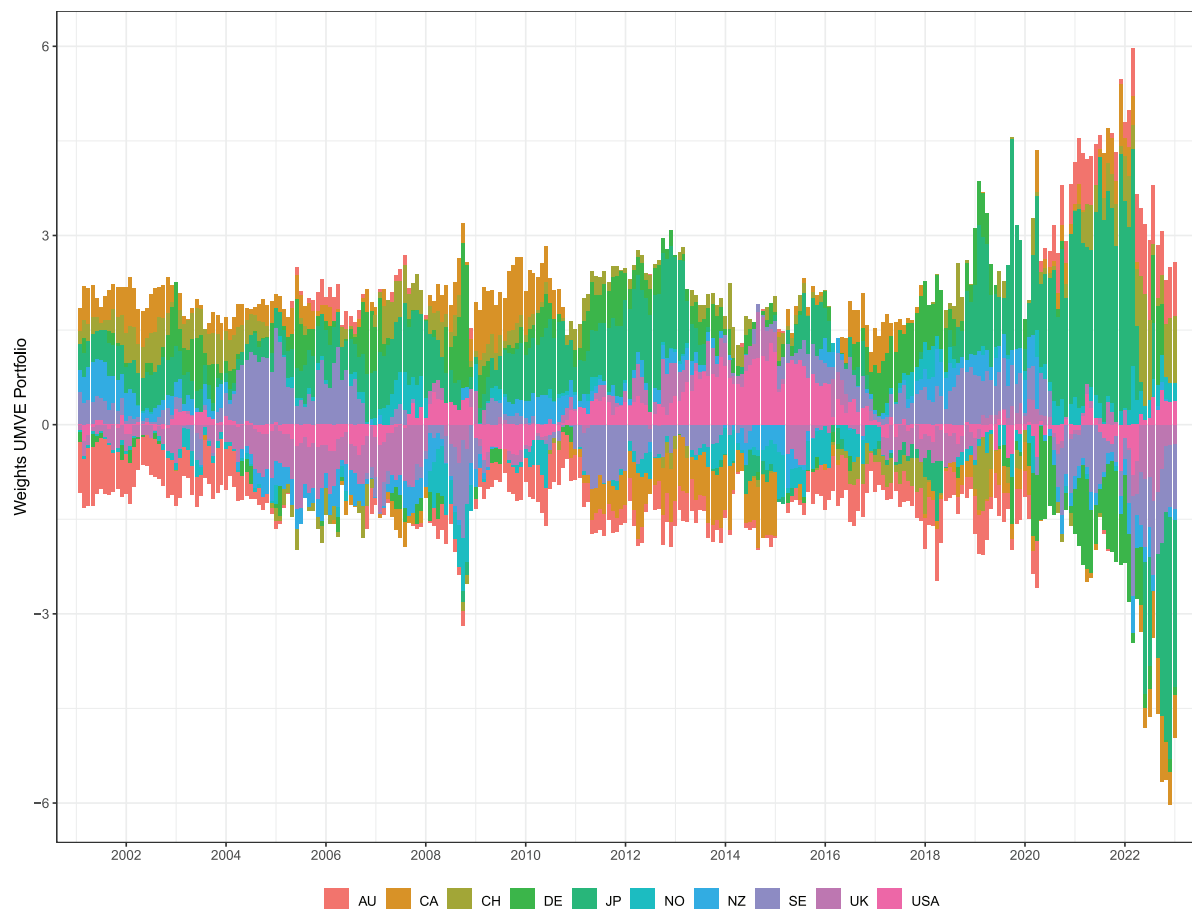


Fig. 9. Times series of portfolio weights.

This figure shows the time series of weights for the UMVE portfolio. The UMVE portfolio returns are scaled so that they have the same volatility as the naive portfolio. Data are from January 2001 to December 2022.

2008, the European sovereign debt crisis from 2010 to 2012, and the recent COVID-19 crisis. The highest values correspond to the fourth quarter of 2022. During that year, average inflation rates reached a record high, triggered also by Russia's invasion of Ukraine in late February 2022. This suggests that the unprecedented sharp increase in yields led by aggressive monetary policy tightening and increases in inflation expectations raised investors' marginal utility of wealth. In accordance with this observation, the difference in the average expected Sharpe ratio between recession and non-recession periods, as defined by the NBER for the U.S. and the Eurostat Business Cycle Clock for the Euro area, is positive and statistically significant.¹⁷ Overall, this suggests that the estimated price of risk is high when the current macroeconomic situation is bleak, consistent with the intuition that investors have high marginal utility in times of crisis and, as a result, require more compensation for foregoing consumption by investing in a risky bond portfolio.

Fig. 10 suggests that the market price of risk in international government bond portfolios is related to the macro economy. This observation aligns with a large literature that links risk premia in single government bond markets to macro variables.¹⁸ Important macro variables that

have been found to affect bond risk premia include output growth, unemployment and inflation (e.g., Ludvigson and Ng, 2009). In a next step, we therefore analyze to what extent expected Sharpe ratios from internationally diversified government bond portfolios are indeed statistically linked to macro variables.

To this end, we aggregate individual countries' macro variables and consider the average trend inflation across these countries (*Countries' trend inflation*), global equity return (*MSCIRet*), as a forward looking measure of global economic growth, and the *VIX* index, as a measure of overall risk, which has been shown by Miranda-Agrippino and Rey (2020) to correlate with the global business cycle. We also include a measure of global economic policy uncertainty, (*GlobEPU*), based on Baker et al. (2016). This measure is computed as the GDP-weighted average of countries' Economic Policy Uncertainty (EPU) indices. Finally, given the globally prominent role of U.S. financial institutions, we also include a measure of the risk bearing capacity of large U.S. institutions, defined by the intermediary capital risk factor (*HKM*) as in He et al. (2017).

If changes in relevant macro variables were perfectly aligned across the G10* countries, macro-driven term-structure changes should also be highly correlated. Consequently, the benefits from allocating capital across different countries' bond markets would be minimal. Diversification matters only when there is heterogeneity in the countries' macroeconomic dynamics. To capture this, we include the cross-sectional standard deviation of the key macro variables (*GDP SD*, *Unempl SD*, *Infla SD*) to explain the dynamics of the UMVE portfolio's expected Sharpe ratio.

Table 13 presents the regression results. Notably, inflation variables emerge as the most consistent and significant determinants of expected

¹⁷ We regress the UMVE expected Sharpe ratio on a constant, plus a recession dummy (equal to 1 for recession periods as defined by the NBER and the Eurostat Business Cycle Clock, 0 otherwise) and document a positive coefficient of 0.260 which is statistically significant at the 5% level, using Newey and West (1994) standard errors.

¹⁸ For recent examples, see Moench and Soofi-Siavash (2022), Fang et al. (2025), or Cieslak and Pflueger (2023).

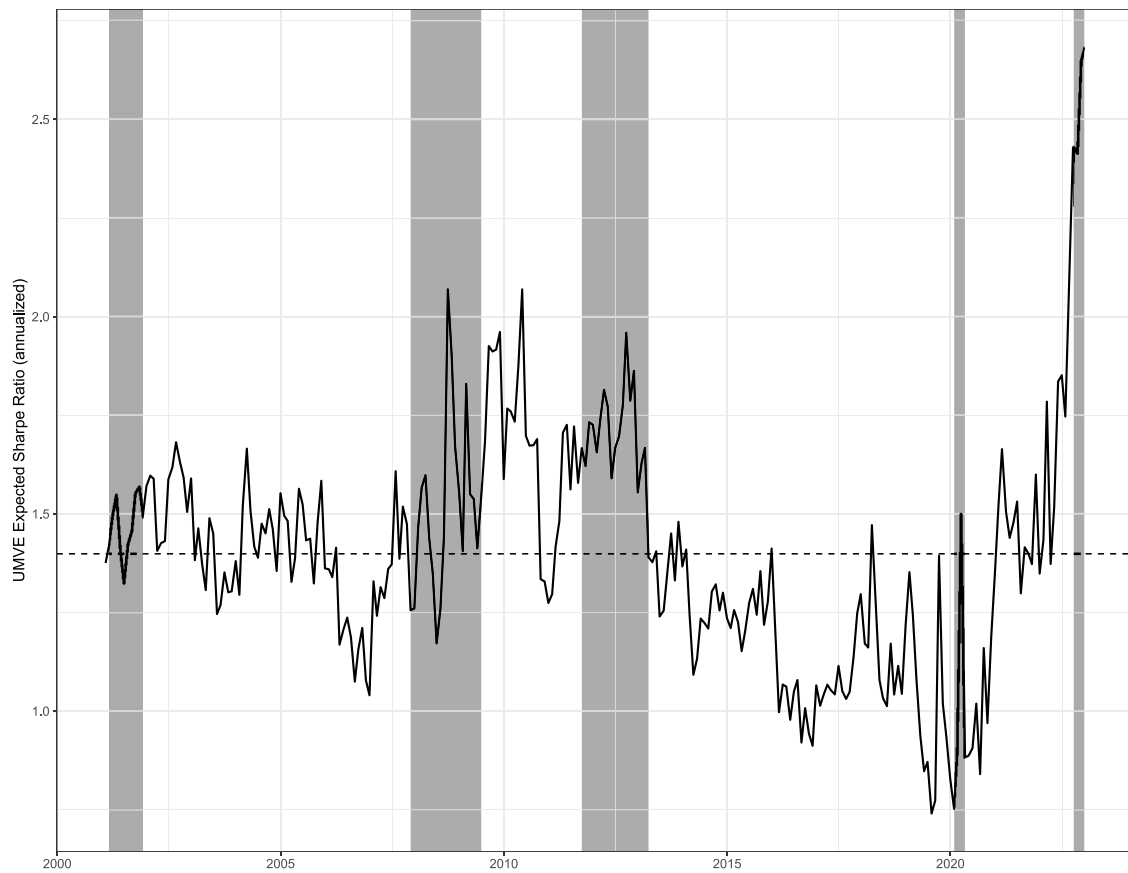


Fig. 10. UMVE expected Sharpe ratio.

The figure shows the time series of the expected Sharpe ratio of the UMVE portfolio (annualized in %). The shaded areas correspond to recession periods, as defined by the NBER for the USA and the Eurostat Business Cycle Clock for the Euro area. The dashed line indicates its mean value. Data are from January 2001 to December 2022.

Table 13

Time series expected UMVE Sharpe ratio and macro and market variables.

	UMVE expected Sharpe ratio			
	(1)	(2)	(3)	(4)
GDP SD	−0.060 (0.040)	0.021 (0.038)	0.020 (0.038)	0.022 (0.038)
Unempl SD	0.006 (0.006)	−0.0001 (0.005)	−0.001 (0.006)	−0.002 (0.006)
Infla SD	0.504*** (0.094)	0.208*** (0.078)	0.203** (0.081)	0.192** (0.085)
Countries' trend inflation		0.856*** (0.137)	0.821*** (0.148)	0.837*** (0.191)
MSCIRet			−0.194 (0.310)	−0.532 (0.376)
VIX Index			0.002 (0.003)	0.002 (0.004)
HKM				0.309 (0.299)
GlobEPU				0.0001 (0.0004)
Constant	0.840*** (0.114)	−0.273 (0.185)	−0.236 (0.198)	−0.256 (0.269)
Observations	264	264	264	264
Adjusted R ²	0.370	0.593	0.594	0.593

This table shows estimates from regressions of the UMVE expected Sharpe ratio on macro and market variables. *GDP SD*, *Unempl SD*, and *Infla SD* are the cross-sectional standard deviations of countries' GDP, unemployment, and inflation rates, respectively. *Countries' trend inflation* refers to the cross-sectional average of countries' trend inflation. *MSCIRet* is the return on the MSCI World Index. *VIX Index* is an index of the implied volatility of the S&P 500 and is calculated from S&P 500 index options. *HKM* is the (He et al., 2017) intermediary capital risk factor, and *GlobEPU* is the Global Economic Policy Uncertainty Index as defined by Baker et al. (2016). Newey–West standard errors with 6 lags are reported in parentheses (Newey and West, 1994). Data are from January 2001 to December 2022. ***, **, * indicates that estimates are statistically significant at the 1, 5, and 10 percent levels, respectively.

Table 14

Time series expected UMVE Sharpe ratio and macro-market variables for the Pre- and Post-08/2008 periods.

	UMVE expected Sharpe ratio	
	Pre-08/2008	Post-08/2008
GDP SD	0.007 (0.099)	0.005 (0.041)
Unempl SD	−0.008 (0.006)	0.003 (0.006)
Infla SD	0.0005 (0.068)	0.253*** (0.093)
Countries' trend inflation	−0.845*** (0.211)	0.873*** (0.196)
MSCIRet	0.275 (0.463)	−1.059** (0.434)
VIX Index	0.019*** (0.004)	−0.001 (0.004)
HKM	0.151 (0.300)	0.462 (0.369)
GlobEPU	0.0003 (0.001)	−0.0001 (0.0004)
Constant	2.534*** (0.341)	−0.310 (0.260)
Observations	91	173
Adjusted R ²	0.322	0.685

This table shows estimates from regressions of the UMVE expected Sharpe ratio on macro and market variables. *GDP SD*, *Unempl SD*, and *Infla SD* are the cross-sectional standard deviations of countries' GDP, unemployment, and inflation rates, respectively. *Countries' trend inflation* refers to the cross-sectional average of countries' trend inflation. *MSCIRet* is the return on the MSCI World Index. *VIX Index* is an index of the implied volatility of the S&P 500 and is calculated from S&P 500 index options. *HKM* is the (He et al., 2017) intermediary capital risk factor, and *GlobEPU* is the Global Economic Policy Uncertainty Index as defined by Baker et al. (2016). Newey–West standard errors with 6 lags are reported in parentheses (Newey and West, 1994). Data for the column labeled *Pre-08/2008* cover the period from January 2001 to July 2008, while data for the column labeled *Post-08/2008* cover the period from August 2008 to December 2022. ***, **, * indicates that estimates are statistically significant at the 1, 5, and 10 percent levels, respectively.

risk adjusted returns. Specifically, over the full sample, a higher cross-sectional standard deviation of inflation and an elevated level of trend inflation are both positively and significantly associated with a higher expected Sharpe ratio in all model specifications.¹⁹

These findings highlight the critical role of inflation as the key driver influencing global government bond markets. This is consistent with Lustig et al. (2019) and Cieslak and Povala (2015), who argue that inflation is important for explaining local currency bond excess returns. The results also accord with recent evidence presented by Andrews et al. (2024), who examine a currency portfolio strategy based on government bond yields. They analyze the excess returns from a portfolio that goes long bonds with the steepest yield curves, financed by 3-month U.S. treasuries, and short bonds of markets with the flattest yields curves, investing the proceeds in 3-month U.S. treasuries. They document that this strategy produces significantly positive excess returns, but only after August 2008, i.e. after the onset of the GFC. They argue that this can be explained by the portfolio's changing exposure to global expected inflation. Given the significant risk-return differences before and after August 2008, we therefore also analyze whether the expected Sharpe ratios of the UMVE portfolio exhibits similar time series differences, and split the sample in a pre- and post-August 2008 period, respectively.²⁰

¹⁹ While inflation emerges as an important explanatory variable, we also investigate whether other macroeconomic variables play a similarly prominent role. Therefore we test the role of alternative macroeconomic indicators: U.S. consumption growth, U.S. employment growth, long-run GDP growth, and sovereign credit risk. Results, which indicate a weak or no relationship, are reported in Appendix D. In untabulated results, we also include the Fama–French 5 factors as additional variables. Results show that these factors are not statistically significantly related to the UMVE expected Sharpe ratio.

4.6.1. Pre and post 2008 periods

Summary statistics for the sub-samples are presented in Table 12. We first observe that the excess returns of the UMVE portfolio analyzed in this paper do not exhibit a structural break around the financial crisis, as their mean values remain quite similar across the two sample periods.²¹ In contrast to Andrews et al. (2024), the Sharpe ratio of the UMVE portfolio in the earlier period is even higher than in the post-crisis period.

However, our results align with Andrews et al. (2024) in the sense that the term-structure slope turns out to be a key predictive variable in both studies. In the context of the UMVE portfolio, this variable has predictive power in both subsample periods, whereas in Andrews et al. (2024) this is only the case in the period after the great financial crisis. This highlights a key difference between a bond portfolio based on term-structure changes, which is funded in the respective local money markets, and a strategy that is fully exposed to exchange rates movements.

Next, we re-run the regression analysis for the two sub-periods and report the results in Table 14. When comparing the subsample results to the ones for the full sample displayed in Table 13, the most important differences are related to the inflation variables. In accordance with Andrews et al. (2024), inflation affects expected Sharpe ratios of the UMVE portfolio differently in the pre- and the post-crisis periods. As shown in Table 14, the dispersion of inflation rates is not significantly associated with the risk premium before 2008, most likely reflecting the generally stable and more homogeneous environment among developed economies in a period sometimes termed the “great moderation”. In contrast, there is a strong positive relationship in the period after the financial crisis.

Table 14 also shows that the level of trend inflation affects expected Sharpe ratios differently before and after August 2008: before, the sign is negative, whereas it is positive afterwards, and in both cases statistically significant. Thus, in the pre-crisis sample period, higher trend inflation is negatively related to expected Sharpe ratios, while the opposite is true in the second subsample period.

These results are consistent with a pre-crisis world, where higher inflation signals higher economic risk, as central banks' inflation targets were frequently exceeded by realized rates. Thus, during this period, a higher trend inflation may have led to a higher demand for precautionary savings, leading to lower expected risk adjusted bond portfolio returns. In contrast, parts of the post-crisis period are characterized by low inflation rates, undershooting central banks' targets and even creating the risk of deflation (see, e.g. Christensen et al., 2015; Fleckenstein et al., 2017). In such a setting, an increase in average trend inflation rates is likely to represent a positive signal about future economic growth, rather than the risk of excessive inflation. In response to an increase in trend inflation, investors are therefore likely to require higher risk adjusted returns for transferring wealth to the future, consistent with higher expected Sharpe ratios.

This finding relates to Andrews et al. (2024), as they find that returns of a slope carry strategy are negative before, but strongly positive after August 2008 and attribute this difference to the strategy's changing exposure to global expected inflation. In contrast, we find that risk-adjusted returns of the UMVE portfolio are not lower before the financial crisis, but, as presented above, inflation is one of the key drivers of the UMVE portfolio's expected Sharpe ratios.²² This highlights the fundamental difference between currency-hedged

²⁰ In untabulated results, we regress the returns of hedged and hedging portfolio strategies on shocks to macroeconomic variables, as well as the intermediary capital factor from He et al. (2017), and MSCI World returns. For most variables, we do not see a consistent pattern. For the priced risk component, shocks to trend inflation and the HKM factor appear to have the highest relevance.

²¹ This is also confirmed by a *t*-test.

Table 15
Regressing bond excess returns on S_{t+1}/S_t .

Country	$rx^{(10Y)}$			$rx^{(10Y),LC}$			$rx^{(10Y),TBill}$		
	Beta	t_beta	adj.R ²	Beta	t_beta	adj.R ²	Beta	t_beta	adj.R ²
Australia	-0.117	-2.649	0.020	-0.119	-2.728	0.021	0.890	20.143	0.579
Canada	-0.129	-2.499	0.017	-0.134	-2.533	0.018	0.873	16.869	0.478
Germany	-0.077	-1.758	0.008	-0.083	-1.898	0.010	0.928	21.101	0.621
Japan	0.035	0.758	0.002	0.033	0.783	0.002	1.037	22.359	0.825
New Zealand	-0.035	-0.751	-0.001	-0.035	-0.755	-0.001	0.969	20.967	0.657
Norway	-0.072	-1.993	0.009	-0.078	-2.106	0.011	0.931	25.338	0.665
Sweden	-0.086	-1.940	0.010	-0.091	-2.067	0.011	0.919	20.436	0.592
Switzerland	0.029	0.851	-0.000	0.021	0.625	-0.002	1.033	29.603	0.766
United Kingdom	-0.140	-2.181	0.018	-0.145	-2.198	0.019	0.862	13.426	0.443

This table shows each country's estimated beta (*Beta*), the corresponding *t*-statistic (*t_beta*), and the adjusted R^2 (*adj.R²*) from regressing bond excess returns on currency depreciation rates S_{t+1}/S_t . We use three definitions of bond excess returns: (i) FX-managed, USD-denominated bond excess returns, as defined in Eq. (6), denoted as $rx^{(10Y)}$; (ii) local currency excess returns, denoted as $rx^{(10Y),LC}$; and (iii) unhedged bond excess returns, computed by borrowing in USD and investing in the foreign long-term bond, denoted as $rx^{(10Y),TBill}$.

portfolios, where bond positions are funded in the local currency, and the slope carry strategy analyzed by Andrews et al. (2024), where bond positions are always funded in the home currency, i.e. the US\$. While beyond the scope of this paper, a detailed analysis of the different role of inflation in FX-risk-hedged versus unhedged bond strategies appears as an interesting direction for future research.

5. The role of exchange rates

The portfolios analyzed in this paper are largely insulated from FX risk, since bond positions are always financed in the local currency. However, excess returns and ultimately UMVE returns may still be related to exchange rates. This section therefore investigates to what extent this is the case and whether any remaining exposure to FX risk is priced.

5.1. International bond returns and exchange rates

We begin by regressing bond excess returns on changes in exchange rates. We hereby consider three alternative definitions of bond returns: (i) bond excess returns as defined in Eq. (6), (ii) local currency excess returns, i.e. returns that would result from a “quanto-position” where the local currency excess return is converted into a USD excess return at a fixed exchange rate; and (iii) FX-unhedged bond excess returns, resulting from borrowing in USD and investing in the foreign bond.

Table 15 presents the results, highlighting two key findings. First, the currency risk exposure of bond excess returns, as defined in Eq. (6), closely mirrors that of local currency excess returns. In both cases, exchange rate changes explain only a marginal proportion of the variation in bond returns, with adjusted R^2 's ranging from 0% to 2%. This contrasts with the regression results for unhedged bond returns, which show a strong relationship with exchange rate movements. In this case, exchange rate changes explain between 44% and 83% of bond excess returns within our sample. Consequently, FX-hedging, as implicit in the construction of bond excess returns defined in Eq. (6), effectively eliminates the majority of FX risk.

²² Andrews et al. (2024) propose an SDF where the heterogeneity across countries is caused by their different exposures to real growth news shocks. In their model, a country's real expected growth can change because expected global growth changes or, indirectly, because expected global inflation changes and this in turn affects expected global growth. The model parameters are calibrated such that shocks to expected global inflation are associated with an increase in the representative agent's marginal utility, whereas the opposite is true for shocks to expected growth. In the pre-crisis period, the slope carry strategy analyzed by Andrews et al. (2024) places more weight on countries which are not very sensitive to global inflation, whereas in the post-crisis period, the strategy shifts portfolio weights to countries with higher exposures to global inflation. As a consequence, required risk premia increase after the crisis.

The second finding from Table 15 is that, while exchange rates only explain a small fraction of the overall variation in FX-risk-hedged bond excess returns, there is still some exposure to currency movements, as evidenced by betas significantly different from zero. However, the economic magnitude of this exposure is minimal, with an average beta of -0.07, comparable to that implicit in local-currency excess returns. In summary, bonds returns as defined in Eq. (6) exhibit only a weak relationship with exchange rate changes.

5.2. PCA's of bond and currency returns

We next conduct a PCA of currency excess returns and compare the resulting loadings with those from FX-risk-hedged bond excess returns. As expected, in both cases, the first principal component (PC) reflects a level factor, with all markets loading positively. However, the correlation between the loadings on the first bond and the first currency PC is low, at 0.16. The second PC of currency excess returns represents a slope (or carry) factor, with positive loadings on high-interest-rate currencies and negative loadings on low-interest-rate currencies. This interpretation does not extend to the second PC of bond excess returns, as discussed in Section 4.4. The correlation between the loadings on the second bond and the second currency PCs is approximately 0.5.

Finally, the third PC lacks a clear economic interpretation, with only modest correlations between the loadings across bonds and currencies. A detailed exposition of this analysis is provided in Appendix H.

In summary, the PCA results highlight that FX-risk-hedged bond returns, as defined in Eq. (6), and currency returns exhibit distinctly different risk structures.

5.3. Bond versus FX UMVE portfolios

Despite the weak relationship between bond returns and exchange rates documented in the previous analyses, an important question remains: is exchange rate risk still reflected in the pricing of FX-risk-hedged international bond portfolios? To address this, we modify the regressions described in Eq. (12) and conduct additional tests to analyze the role of exchange rate risk.

Specifically, we orthogonalize the UMVE returns, rx^U , by regressing them on a currency SDF, rx^{FX} . The resulting error term represents a pure bond SDF, $rx^{U\perp}$, which is determined only by term structure movements unpriced by the currency SDF. We obtain the currency SDF defined by the unconditionally mean-variance-efficient currency portfolio return, FX-UMVE, constructed from the currencies in the G10 markets, from Chernov et al. (2024).²³

Chernov et al. (2023) demonstrate that the FX-UMVE effectively prices both individual currency returns and dynamic currency trading

²³ We are very grateful to Mike Chernov for sharing the data for the FX-UMVE.

Table 16

Testing bond and FX UMVEs using individual country bond returns.

Country	$rx^{U\perp}$			rx^{FX}			rx^U and rx^{FX}		
	Alpha	t_alpha	adj.R ²	Alpha	t_alpha	adj.R ²	Alpha	t_alpha	adj.R ²
Australia	0.882	0.465	0.012	2.070	1.180	−0.003	0.837	0.446	0.009
Canada	0.683	0.442	0.085	3.178	2.185	−0.004	0.743	0.490	0.082
Germany	1.030	0.800	0.077	3.181	2.192	0.002	0.853	0.654	0.080
Japan	0.559	0.798	0.171	2.233	3.051	−0.003	0.599	0.831	0.168
New Zealand	0.470	0.263	0.032	2.108	1.265	−0.001	0.325	0.183	0.031
Norway	1.641	1.047	0.012	2.452	1.649	0.003	1.391	0.902	0.015
Sweden	1.216	0.834	0.050	2.954	1.921	0.004	0.979	0.660	0.054
Switzerland	0.858	0.722	0.080	2.739	2.278	−0.001	0.782	0.666	0.080
United Kingdom	0.348	0.209	0.044	2.531	1.481	−0.004	0.441	0.261	0.041
United States	0.816	0.412	0.078	3.567	1.927	0.006	0.487	0.246	0.085

This table shows the alpha from regressing countries' excess returns on alternative UMVE portfolio returns (*Alpha*, in % p.a.), the corresponding *t*-statistics (*t_alpha*), and the adjusted R² from each regression. We consider the following alternative UMVE portfolio returns: i) $rx^{U\perp}$, the orthogonalized bond UMVE portfolio, whose returns are uncorrelated with currency risk; ii) rx^{FX} , the UMVE portfolio constructed from currency excess returns; and iii) a combined specification using both rx^U (the UMVE portfolio based on bond excess returns) and rx^{FX} .

Table 17

Testing bond and FX UMVEs using portfolio strategies.

Strategy	$rx^{U\perp}$			rx^{FX}			rx^U and rx^{FX}		
	Alpha	t_alpha	adj.R ²	Alpha	t_alpha	adj.R ²	Alpha	t_alpha	adj.R ²
1/ <i>N</i>	0.850	0.679	0.071	2.701	2.148	0.000	0.744	0.600	0.071
GDP-weighted	0.769	0.552	0.092	3.062	2.212	0.003	0.601	0.431	0.096
Average Carry	1.265	1.014	0.107	3.665	2.938	−0.003	1.293	1.049	0.104
CS-Carry	0.813	1.033	0.209	2.928	3.463	−0.002	0.827	1.043	0.208
TS-Carry	0.738	0.676	0.091	2.652	2.399	−0.002	0.703	0.653	0.089
Average Value	0.244	0.193	0.035	1.891	1.494	−0.002	0.462	0.358	0.034
CS-Value	0.881	1.273	0.014	1.347	1.942	−0.002	0.831	1.196	0.012
TS-Value	0.788	0.763	0.058	2.166	2.147	−0.002	0.736	0.734	0.057
Average Mom12	1.244	0.964	0.057	2.922	2.327	−0.001	1.152	0.892	0.057
CS-Mom12	0.810	1.024	−0.004	0.897	1.277	−0.002	0.879	1.123	−0.006
TS-Mom12	1.222	1.128	0.040	2.539	2.387	−0.004	1.291	1.208	0.037
Average Mom1	1.254	0.969	0.008	0.292	0.230	−0.002	1.085	0.836	0.006
CS-Mom1	−1.446	−1.830	0.001	−1.221	−1.640	−0.003	−1.496	−1.894	−0.002
TS-Mom1	0.646	0.645	0.008	0.019	0.020	−0.004	0.631	0.626	0.004

This table shows the alpha from regressing bond trading strategy returns on alternative UMVE portfolio returns (*Alpha*, in % p.a.), the corresponding *t*-statistics (*t_alpha*), and the adjusted R² from each regression. We consider the following alternative UMVE portfolio returns: i) $rx^{U\perp}$, the orthogonalized bond UMVE portfolio, whose returns are uncorrelated with currency risk; ii) rx^{FX} , the UMVE portfolio constructed from currency excess returns; and iii) a combined specification using both rx^U (the UMVE portfolio based on bond excess returns) and rx^{FX} .

Table 18

Sharpe ratio and GRS test for bond and FX UMVEs using portfolio strategies.

Model	SR	GRS Tests — p-values			
	UMVE	All strategies	All Carry	All Value	All Carry and Value
rx^U (the original UMVE)	1.229	0.455	0.399	0.635	0.544
$rx^{U\perp}$	1.236	0.461	0.415	0.607	0.544
rx^{FX}		0.004	0.000	0.124	0.001
$rx^U + rx^{FX}$		0.439	0.356	0.709	0.551

This table shows in column (1) the Sharpe ratio (SR) of the UMVE portfolios. Column (2) shows the *p*-value of the GRS test if the alphas from regressing all global portfolio strategies on the respective UMVE portfolio are jointly equal to zero. Column (3) shows the *p*-value of the GRS test if the alphas from regressing global carry portfolio strategies on the respective UMVE portfolio are jointly equal to zero. Column (4) shows the *p*-value of the GRS test of whether the alphas from regressing global value portfolio strategies on the respective UMVE portfolio are jointly equal to zero. Column (5) shows the *p*-value of the GRS test of whether the alphas from regressing global carry and value portfolio strategies on the respective UMVE portfolio are jointly equal to zero. Data are from January 2001 until December 2022.

strategies. To orthogonalize the returns of the bond UMVE portfolio with respect to the currency SDF, we regress the bond UMVE returns on the demeaned FX UMVE returns, as shown in Eq. (15):

$$rx^U = \alpha + \beta(rx^{FX} - \overline{rx^{FX}}) + \epsilon \quad (15)$$

To obtain mean-preserving orthogonalized UMVE portfolio returns, $rx^{U\perp}$, we define $rx^{U\perp} = \alpha + \epsilon$. Using $rx^{U\perp}$ and rx^{FX} , we perform asset pricing tests akin to those reported in Section 4.2. Results for the two sets of test assets are reported in Tables 16 (countries) and 17 (strategies). We find that the component of the UMVE portfolio returns unrelated to the currency SDF, $rx^{U\perp}$, effectively prices both individual countries' bond returns and dynamic bond portfolio strategies. Indeed,

the resulting alphas, shown in the columns to the left, are not statistically different from zero, as all *t*-statistics are low, with the largest one being −1.8 for CS-Mom1. Our evidence thus suggests that taking out priced currency risks does not materially affect the UMVE's ability to price international bond markets and portfolio strategies.

Next, we perform similar tests, but use the currency SDF, rx^{FX} , as the explanatory variable in the regressions. The results are reported in the middle columns of Tables 16 and 17. In this case, most alphas are large and statistically different from zero. Thus, the currency SDF does a poor job in pricing FX-risk-hedged international bond portfolios.

We also examine the pricing performance of the combination of rx^U (the original UMVE return) and rx^{FX} (the FX-UMVE). Analyzing the three rightmost columns of Tables 16 and 17, we observe that including

the currency SDF as an additional explanatory variable has minimal impact on the ability to price bonds or bond strategies. The alphas and their t -statistics remain nearly identical to the results presented earlier in Tables 6 and 7.

We conduct GRS tests to determine whether the alphas from different model specifications are jointly different from zero. The results in Table 18 indicate negligible differences across models, whether using rx^U , $rx^{U\perp}$ or a combination of rx^U and rx^{FX} . These results accord well with our earlier evidence, suggesting that exchange risk contributes minimal explanatory power to bond excess returns, as defined in Eq. (6).

When relating the bond UMVE and the FX UMVE to the macroeconomy, we find distinct patterns. While the expected Sharpe ratios of the former are primarily related to inflation dynamics across countries (see Section 4.6), Chernov et al. (2023) find that FX UMVE returns are primarily related to US consumption growth. The important role of inflation for the bond UMVE is intuitive. Excess returns of long-term bonds are almost mechanically linked to inflation shocks, as inflation erodes the purchasing power of fixed future cash flows and correlates with the level of interest rates. Consequently, it is natural for inflation risk to be priced in international bond markets. This is consistent with Fang et al. (2025), who show that government bonds have negative exposure to both core and energy inflation. In contrast, these authors find that currencies behave differently: While they exhibit positive and sizable exposure to energy inflation, exposure to core inflation is negative, albeit not statistically significant. This fact likely reflects the tendency of central banks to intervene – either directly in currency markets or indirectly through monetary policy decisions – thereby mitigating some of the effects of core inflation shocks on exchange rates. Thus, while Fang et al. (2025) find that certain FX trading strategies – such as carry – are exposed to inflation, the opposing signs of currencies' exposures to core and energy inflation, which tend to offset each other, may explain why headline inflation has not been found to be significantly related to the FX UMVE.

6. Implications for bond portfolio strategies

In this section, we discuss implications of our findings for bond portfolio management. We start by illustrating the time-varying expected risk premia of traditional bond portfolio strategies. We then provide information about the time-varying composition of the UMVE portfolio, and discuss to what extent extreme long and short positions, as shown in Table 11, are required to obtain the attractive risk-return relation of the UMVE.

Once the portfolio projection of the SDF is known, one can calculate the expected risk premium of various bond strategies in real time. Specifically, for each bond strategy considered by an investor, she can obtain the expected Sharpe ratio based on estimates of expected excess returns and the covariance matrix. Fig. 11 illustrates the cross-sectional and the time variation of expected Sharpe ratios for the trading strategies described in Section 4.1. Positive (negative) values are shown in green (red) color; darker areas indicate larger absolute values. First, Fig. 11 reveals considerable heterogeneity: simple diversification (naive and GDP-weighted), and carry strategies have almost always positive expected Sharpe ratios, while momentum strategies show notable fluctuations between positive and negative values. The attractiveness of value strategies appears to vary at lower frequency.

While a real-time assessment of the risk-return properties of a given trading strategy, as illustrated in Fig. 11, may be useful for investors, it does not speak to the overall optimal portfolio. We thus turn to the analysis of the practical implementability of the UMVE portfolio. We have shown that hedged portfolio strategies lead to economically attractive risk-return ratios which, in most cases, beat the original trading strategies by a wide margin. It is therefore natural to ask whether these strategies would be implementable in practice. Such a practical implementation may face two challenges. First, the hedged portfolio

strategies may require extreme position weights to achieve their high performance. To see how sensitive the results are with regard to a portfolio manager's ability to take extreme market positions, we impose minimum and maximum weights for the hedged portfolio strategies and compare the resulting Sharpe ratios with the ones of the original strategies. A second potential concern relates to the feasibility of taking short positions. In principle, short positions should be easily achievable, since long-term government bond futures are available in all markets of our main sample with the exception of Norway.²⁴ However, many asset managers, such as conventional mutual fund managers, face legal or investor-imposed short-selling constraints. We therefore also analyze strategies with minimum weights of zero in each market as a robustness check. Specifically, we implement the following two lower and upper bounds on portfolio weights: First, we restrict portfolio weights for each country between 0% and 20% and second, we consider a restricted strategy with portfolio weights in the range between −20% to +20%. We report the results in Fig. 12. Remarkably, even when we apply such relatively conservative weight restrictions, the hedged strategies beat the original trading strategies with Sharpe ratios close to those achieved by the unrestricted portfolios. However, it appears that a considerable part of the performance comes from short positions. While 13 out of 14 long-only strategies still outperform their unhedged counterparts, their Sharpe ratios tend to be lower than those of portfolios that allow at least limited short positions. This observation does not hold for the momentum portfolios. Here, restricting investors from taking short positions even turns out to be beneficial. Overall, the hedged portfolios improve Sharpe ratios considerably, even when large portfolio weights are not allowed.

Finally, Table 11 shows that implementation of the UMVE requires high portfolio turnover which might raise concerns about trading costs. Yet, this is not likely to be an important issue for major government bond markets since they all feature derivatives markets. Trading costs for bond futures are very low, typically in the order of magnitude of 1 basis point. Transactions in the cash market tend to be slightly more expensive. Favero et al. (2010) quantify the median cost of trading German government bonds with 10 year maturity with 3 basis points. Using a range from 1 to 3 basis points, trading costs will lower the return of the UMVE by only about 0.2% to 0.6% p.a.

7. Robustness

7.1. Estimation of expected returns: Alternative specifications

In this subsection, we test alternative specifications of Eq. (7). Specifically, we (i) add a constant, (ii) add country fixed effects, and use alternative definitions of (iii) the forward spread and (iv) the bond value variables.

For alternative specification (iii), we compute the forward spread variable defined over 1 year:

$$ForwardSpread_t^* = \frac{P_{i,t}^{(9Y)}}{P_{i,t}^{(10Y)}} - \frac{1}{P_{i,t}^{(1Y)}}, \quad (16)$$

and not over 1 month, as defined in Eq. (8).

²⁴ Kojien et al. (2018) include the same ten bond markets as our paper but calculate synthetic bond futures returns because of liquidity concerns and data availability. For markets where they have access to bond futures prices, they find correlations above 0.95 between actual and synthetic futures returns. Similarly, we find correlations above 0.97 between excess returns that we calculate from yield curves and futures excess returns for Australia, Canada, Germany, Sweden, Switzerland, the UK, and the U.S. We find somewhat lower correlations for futures markets that are less liquid (New Zealand 0.81, Japan 0.92). There is no long-term government bond futures contract traded for Norway.

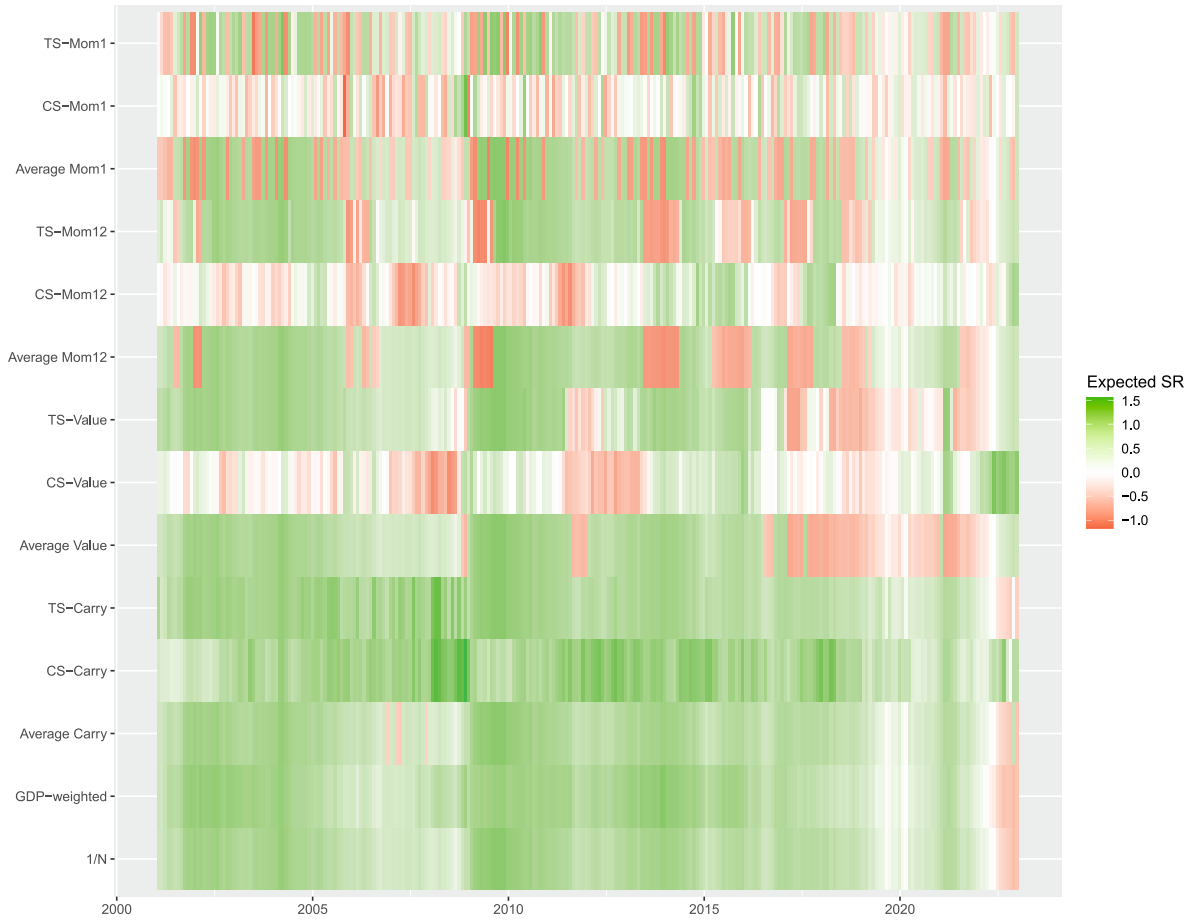


Fig. 11. Portfolio strategies heatmap.

This figure shows a heatmap of the portfolio strategies' expected Sharpe ratio over time. The estimation of the conditional means and the conditional covariances follows Sections 3.4.1 and 3.4.2. Detailed description of the trading strategies is available in Table 5. Data are from January 2001 to December 2022.

The alternative bond value variable used in specification (iv) is based on trend inflation, as defined by Cieslak and Povala (2015), instead of historical inflation. Specifically, we compute:

$$Value_t^* = Y_{i,t}^{(10Y)} - \tau_t^{CPI} \quad (17)$$

$$\tau_t^{CPI} = \frac{(1-v)}{(1-v^M)} \sum_{i=0}^{M-1} v^i \pi_{t-i},$$

where $\pi_t = CPI_t / CPI_{t-12M} - 1$ is the annual inflation in month t , lagged by one month, $M = 120$ is the number of months considered for measuring inflation²⁵, and v is a parameter calibrated to inflation survey data which we set equal to 0.987, as in Cieslak and Povala (2015). Trend inflation, as defined above, has been shown to forecast future inflation and to predict bond excess returns (see Cieslak and Povala (2015) and the literature cited there).

The results for alternative definitions of return prediction variables (i) to (iv) are reported in Appendix D. These results closely align with those obtained from the main model specification, but do not outperform it.

7.2. Estimation of expected returns: Country-by-country dynamics

We estimate the predictive regression specified in Eq. (7) using pooled OLS, which provides a single estimate for $\hat{\gamma}_t$ and $\hat{\delta}_{i,t}$, respectively.

²⁵ When fewer than 120 months of data are available, we progressively adjust M to ensure the same number of observations as in the main model.

While this is efficient if the underlying relationships are expected to be similar or identical across countries, it might bias results if there are cross-country differences. As a robustness check, we therefore run country-by-country regressions over expanding time windows, allowing coefficients $\hat{\gamma}_{i,t}$ and $\hat{\delta}_{i,t}$ to vary across countries i and time t . Appendix H presents the resulting time series of gamma and delta coefficients for each G10* market, as well as cross-sectional averages and standard deviations. Compared to the pooled OLS estimation, individual countries' estimates are more unstable at the beginning of the sample, with much wider confidence intervals. While the point estimates of coefficients vary across countries and time, the general pattern is remarkably similar, and the cross-sectional mean of the individual countries' estimates is close to the estimate when using the pooled OLS approach.

7.3. Estimation of expected returns: Including additional predictive variables

We next explore the robustness of our findings with respect to the variables used in the model of expected returns. In particular, we consider alternative versions of momentum, foreign exchange predictors, additional forward interest rates, as suggested by Cochrane and Piazzesi (2005), and a credit risk measure as additional candidate variables.

7.3.1. Long and short term momentum

We start by extending the model in Eq. (7) and include long-term and short-term bond excess return momentum. Specifically, we

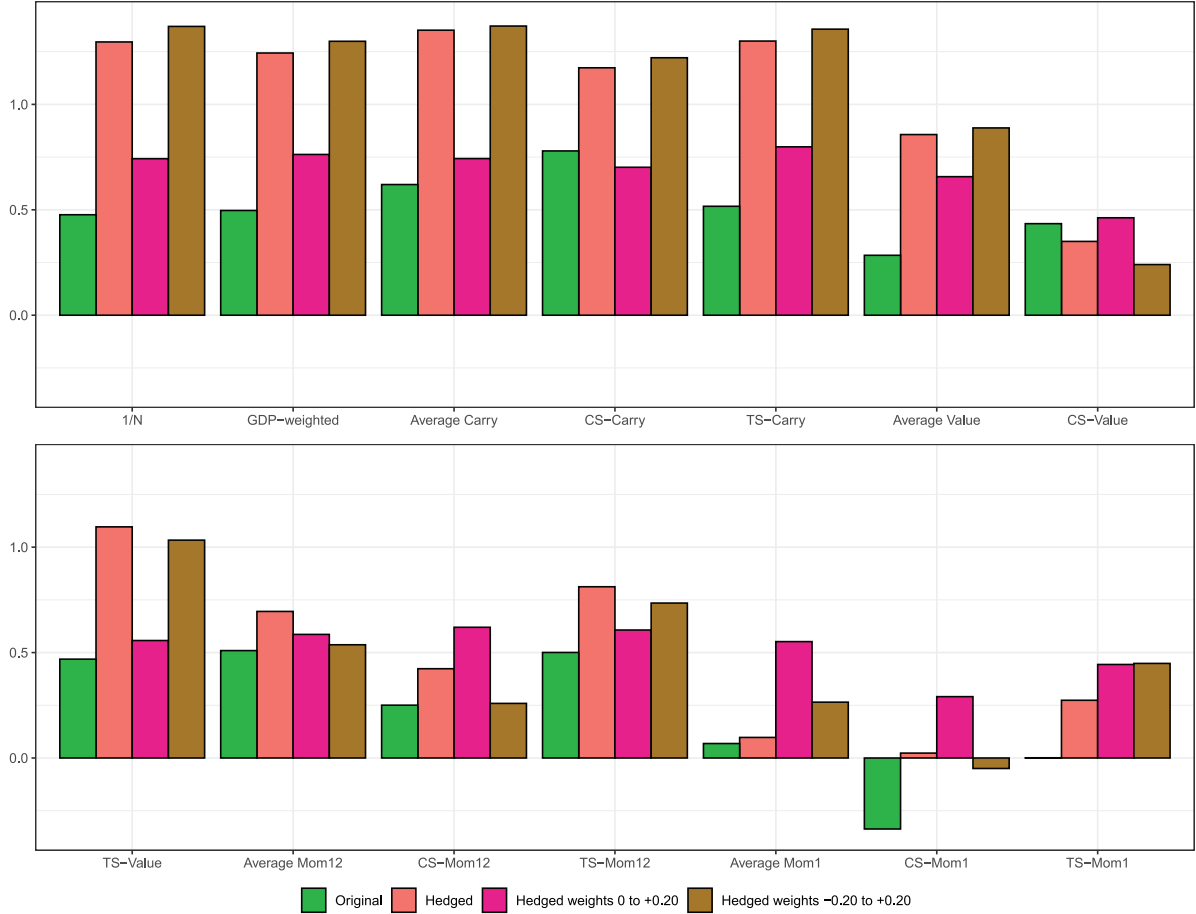


Fig. 12. Portfolio strategies with weights cutoffs.

This figure shows the annualized Sharpe ratio of the original, hedged portfolio strategies, and hedged portfolio strategies with weight restrictions. Hedged weights 0 to +0.20 means that the weights of the hedged portfolio are restricted to be between 0% and 20% of the total portfolio exposure. Hedged weights −0.20 to +0.20 means that the weights of the hedged portfolio are restricted to be between −20% and +20% of the total portfolio exposure.

estimate the following out-of-sample regressions:

$$\mathbb{E}_t(rx_{t+1}^i) = \psi_{1,t} Forward Spread_t^i + \psi_{2,t} Value_t^i + \psi_{3,t} LT Mom_t^i + \psi_{4,t} ST Mom_t^i, \quad (18)$$

where *Forward Spread* and *Value* are defined as in Section 3.4.1, *LT Mom* represents the cumulative return from $t - 12$ to $t - 1$ and *ST Mom* denotes the most recent monthly return. We find that incorporating these additional variables weakens the results compared to the model that includes only carry and value. Specifically, this alternative model produces noisier conditional expectations. Importantly, although a UMVE portfolio constructed from the extended model above still seems to correctly price all trading strategies, its Sharpe ratio is lower than the one resulting from the UMVE that only relies on two signals. Detailed results are provided in Appendix D.

7.3.2. Foreign exchange predictors

As detailed in Section 5, the FX-risk-hedged returns we use are largely insulated from exchange rate risk. However, it could be argued that foreign exchange predictors still play a role in forecasting excess bond returns. To investigate this, we use the FX predictors proposed by Chernov et al. (2023) and estimate the following out-of-sample regressions:

$$\begin{aligned} \mathbb{E}_t(rx_{t+1}^i) = & b_{1,t} Forward Spread_t^i + b_{2,t} Value_t^i \\ & + b_{3,t} (S_t^i / F_t^i - 1)_t + b_{4,t} z_t^i + b_{5,t} (S_t^i / S_{t-12}^i - 1), \end{aligned} \quad (19)$$

where $(S_t^i / F_t^i - 1)$ is the forward discount, z_t^i is the Real Exchange Rate (RER)²⁶, and $(S_t^i / S_{t-12}^i - 1)$ is a trend signal.

Adding foreign exchange predictors results in weaker predictability.²⁷ Additionally, the resulting UMVE fails to price several test assets such as Japanese bonds and some strategies related to carry and momentum. We report detailed results in Appendix D.

7.3.3. Additional forward interest rates (Cochrane and Piazzesi)

Cochrane and Piazzesi (2005) demonstrate that a linear combination of forward rates has significant predictive ability to explain future excess bond returns. Motivated by this finding, we analyze a model of expected returns by replacing the forward spread in Eq. (7) with a linear combination of forward rates, incorporating maturities of up to 5 years. Specifically, the regression equation is specified as follows:

$$\mathbb{E}_t(rx_{t+1}^i) = v_{1,t} Value_t^i + v_{2,t} Y_t^{(1Y),i} + v_{3,t} F_t^{(2Y),i} + v_{4,t} F_t^{(3Y),i} + v_{4,t} F_t^{(4Y),i} + v_{4,t} F_t^{(5Y),i}, \quad (20)$$

²⁶ First define the RER as $Q_t^i = S_t^i \cdot CPI_t^{US} / CPI_t^i$. Then, divide each RER by the average RER from 4.5 to 5.5 years ago: $\tilde{Q}_t^i = Q_t^i \left(\frac{1}{13} \sum_{j=-6}^6 Q_{t-60+j}^i \right)^{-1}$. Finally, demean cross-sectionally: $z_t^i = \tilde{Q}_t^i - \frac{1}{N} \sum_{i=1}^N \tilde{Q}_t^i$.

²⁷ This is confirmed by the Diebold and Mariano (1995) test comparing the full model that includes bond and value signals with the one that additionally includes foreign exchange predictors. The test shows a significantly higher predictive ability for the full model, with a t -statistic of −6.3.

where $F_t^{(m)}$ is defined as $P_t^{(m-1)}/P_t^{(m)}$. Panel C of Table D3 in the appendix reveals that the unbiasedness test using the expected returns estimated from Eq. (20) gives a weaker result compared to the baseline optimal model. Consequently, adding a linear combination of forward rates does not improve the estimation of bond expected returns.

7.3.4. Credit risk

The G10* countries, as defined in Section 3.1, all represent developed economies that are both economically and politically stable. Consequently, we interpret their bond markets as effectively default-free. However, existing literature highlights credit risk as a potentially important component of the bond risk premium, even for such countries (Longstaff et al., 2011; Augustin et al., 2022). To address this, we perform an additional robustness check by incorporating sovereign credit risk, measured using sovereign CDS spreads, into the estimation of expected returns.

Specifically, we run out-of-sample regressions of the following form:

$$\mathbb{E}_t(r_{t+1}^i) = \theta_{1,t} \text{Forward Spread}_t^i + \theta_{2,t} \text{Value}_t^i + \theta_{3,t} \Delta \text{Credit Risk}_t^i, \quad (21)$$

where $\Delta \text{Credit Risk}_t^i$ is computed as $\text{CDS}_{i,t} - \text{CDS}_{i,t-1M}$, with $\text{CDS}_{i,t}$ and $\text{CDS}_{i,t-1M}$ being the monthly average CDS spread levels (in basis points) at time t and $t - 1M$, respectively. We use data on 5-year USD-denominated sovereign CDS spreads from Markit and follow Della Corte et al. (2022) for an accurate selection of the CDS restructuring clauses.²⁸ To obtain a balanced panel of liquid CDS contracts, we use data starting from the end of 2003. However, we lack liquid data for Switzerland (up to 2005) and the United Kingdom (up to 2008). To proxy for the missing CDS data, we use contracts of comparable countries based on credit ratings and levels of investor protection.²⁹ We find that adding CDS does not improve the prediction of bond expected returns or variances. We provide results in Appendix D.

7.4. Estimation of expected returns: Fixed estimation window

In the main analysis, expected returns are estimated using an expanding time window. However, using a rolling estimation window may be more effective in capturing potential time variation in the relationship between predictive variables and future bond excess returns. To explore this, we re-estimate Equation (7) using fixed 5-year rolling windows instead of an expanding window.

Results remain broadly unchanged. As expected, the time series of the estimated coefficients $\hat{\gamma}_t$ and $\hat{\delta}_t$ show greater fluctuation. Nevertheless, the estimates of expected returns remain valid and unbiased. We document the output of these robustness tests in Appendix D.

7.5. Regression test of the UMVE portfolio: Other maturities

We have demonstrated that the UMVE derived in Sections 2 and 3 effectively prices individual bond markets and trading strategies. These tests were based on holding-period returns from 10-year bonds. To analyze whether the UMVE also prices international bonds in other maturity buckets, we extend the analysis to include holding-period returns

from alternative bond maturities. Specifically, we analyze whether the UMVE also prices 5-year and 15-year maturities.

Results are reported in Table D8 in the appendix. Notably, none of the regressions yields a statistically significant alpha at the conventional 5% level. This finding holds even for the shorter 5-year maturity, where one might expect distortions or structural breaks due to countries' monetary policies following the Great Financial Crisis and COVID-19 — factors that may not be fully captured by the 10-year return-based UMVE. Overall, this evidence further reinforces the validity of the UMVE.

7.6. Alternative measurement currency

The analysis in this paper is based on bond excess returns denominated in USD. This raises the question whether the results also hold for alternative measurement currencies. To address this, we follow Chernov et al. (2024) and convert the USD-based UMVE into alternative measurement currencies. For instance, when focusing on EUR, we compute $r_{t+1}^{U, EUR} = r_{t+1}^U \cdot S_t^{EUR} / S_{t+1M}^{EUR}$, where S_t^{EUR} represents the exchange rate in USD per unit of EUR. We then examine whether $r_{t+1}^{U, EUR}$ prices individual bond returns and strategies, expressed in EUR. We perform this exercise for all currencies in our sample.

As detailed in Appendix I, changing the measurement currency does not qualitatively alter the findings. SDFs constructed from alternative measurement currencies consistently price the respective bond portfolio returns effectively. This result aligns with the discussion in Section 5, which shows that the exposure of excess bond returns to FX risk is economically very small. Consequently, the choice of measurement currency used to define the SDF does not materially affect the conclusions.

7.7. Alternative benchmark portfolios

7.7.1. The naive portfolio

As shown above, the SDF projection on the UMVE portfolio is supported by standard validation tests. However, one may question whether a naive SDF, such as the one derived from a projection on the $1/N$ portfolio, would yield similar results. To investigate this, we conduct regression tests analogous to those in Tables 6 and 7, substituting the naive $1/N$ portfolio returns when defining the SDF. The results are presented in Appendix D.

While the $1/N$ portfolio successfully prices some individual government bond markets and portfolio strategies, we observe that, on average, the t -statistics of the alphas are significantly higher than those obtained when using the SDF based on the UMVE portfolio (see Tables 6 and 7). More critically, for certain test assets, the alphas resulting from the $1/N$ portfolio are statistically significantly different from zero. These findings lead us to conclude that the SDF derived from the naive $1/N$ portfolio strategy is dominated by the SDF derived in Section 2.

7.7.2. The minimum entropy approach

In our main analysis, we obtain real-time estimates of the UMVE-based SDF. An alternative approach is based on minimum entropy, which takes into account higher moments of returns. Following Sandulescu et al. (2021), we provide an in-sample estimate of the minimum entropy SDF for international government bond markets. Specifically, we obtain the minimum entropy SDF, $\tilde{M}$, as

$$\tilde{M}_t = R_{\hat{\lambda}^*, t}^{-1}, \quad (22)$$

where $R_{\hat{\lambda}^*, t}$ denotes the gross returns of the optimal growth portfolio. The latter is defined as

$$R_{\hat{\lambda}^*, t} = \sum_{i=1}^N \hat{\lambda}_i^* r_{t+1}^i + R_{US}^{(1M)}, \quad (23)$$

where $\hat{\lambda}_i^*$ is the optimal portfolio weight for country i and $R_{US}^{(1M)}$ is the gross return of 1-month U.S. T-bills. We estimate the optimal portfolio

²⁸ We focus on contracts denominated in USD and with 5-year maturity, as they are the most liquid contracts in the sovereign CDS market.

²⁹ To identify comparable countries, we match credit ratings, using Thomson Reuters as data source for Fitch Long Term Default Ratings. To match investor protection, we use La Porta et al. (1998) and select countries of the same legal origin and with a low number of deviations in all sub-categories of shareholder rights, creditor rights, and rule of law. For these countries, we compute the average CDS spreads and use them for the period for which the respective matched country has no liquid CDS data. To avoid potential jumps between the two time series, we apply an adjustment factor to the matched average CDS spread.

weights from the solution to the following set of empirical moment conditions:³⁰

$$\hat{\mathbb{E}}[R_{\lambda^*}^{-1} \mathbf{r}\mathbf{x}] = \mathbf{0}, \quad (24)$$

where $\mathbf{r}\mathbf{x}$ is the vector of excess returns of our test assets, i.e. the individual bond markets. Following Sandulescu et al. (2021), we obtain the vector of portfolio weights λ^* , i.e. the parameters to be estimated from Eq. (24), using the method of moments.

We report the results of the hypothesis test specified in Eq. (12) and the results from the GRS test in Appendix D. Overall, the optimal growth portfolio, which is constructed in-sample, explains a larger fraction of the variation in bond returns and portfolio strategies than the UMVE portfolio, which is constructed out-of-sample. However, the UMVE portfolio dominates with respect to pricing international government bonds. With the exception of cross-sectional carry, cross-sectional value, and cross-sectional short-term momentum, alphas of individual country and strategy regressions are statistically indistinguishable from zero. However, using the GRS test, we reject the hypothesis that alphas are jointly equal to 0 when using the minimum entropy SDF.

8. Conclusion

This paper investigates the pricing of currency-hedged government bond portfolios and presents several key findings. We document substantial diversification benefits for investors due to the imperfect correlation between term-structure movements across bond markets, even when focusing only on developed markets and simple strategies, such as GDP-weighted or equally weighted portfolios.

We find that the unconditional mean–variance efficient portfolio of G10* government bonds can be constructed using expected return forecasts based on forward spreads and real yields, along with a time-varying covariance matrix, estimated with a shrinkage method. The resulting SDF prices individual G10* bond markets and dynamic bond trading strategies effectively. Although the SDF derived in this paper is based on 10-year bond portfolios, we demonstrate that it also prices bonds of other maturities.

We explore several model extensions, incorporating additional variables into the expected return models, such as CDS-based sovereign credit risk measures, the term structure of forward interest rates and alternative definitions of carry and value. We also confirm the results using alternative measurement currencies. Remarkably, the parsimonious UMVE model from the main analysis prices this asset class surprisingly well, even when compared to these more complex extensions.

We also investigate to what extent the bond pricing kernel captures FX risk. First, we demonstrate that, as bond positions are financed in their respective local currencies, the UMVE portfolio's exposure to FX risk is economically negligible. Moreover, we show that the returns of the UMVE portfolio are largely unrelated to an SDF that prices currencies. Consequently, using only the component of the UMVE return that is orthogonal to the currency SDF prices international bond portfolio equally well as the original UMVE. These findings highlight that the priced risks of international bond portfolios are fundamentally distinct from those of currency portfolios.

The UMVE portfolio has several notable properties. First, the optimal international government bond portfolio achieves a large improvement in the Sharpe ratio, increasing it from an average of approximately 0.46 for individual markets to a value exceeding 1. However, the UMVE exhibits substantial time-variation in the implied market price of risk. Specifically, it increases around recessions and crises, with the highest expected Sharpe ratio observed in the fourth quarter of 2022, when concerns about inflation and tightening monetary policies raised investors' marginal utility of wealth.

Additionally, the paper finds that the expected Sharpe ratio of the UMVE portfolio is strongly positively related to the absolute level and to the dispersion of inflation rates across the G10* countries. These relationships seem to be largely driven by the period after August 2008.

While bond returns exhibit a strong factor structure, with 86% of the variance being explained by the first three principal components, there is very little relation between common sources of variation in international bond market returns and priced risk. The first 10 principal components only explain around 30% of the UMVE portfolio return variation.

Consistent with this finding, portfolio strategies, such as equally weighting or various factor-based portfolio strategies, such as carry, value, and momentum strategies, all exhibit large amounts of unpriced risks. By constructing portfolios that hedge out such unpriced components from bond strategies, Sharpe ratios improve substantially. Our analysis also suggests that, for successful practical implementation, the estimation of expected returns is essential, while estimation of variances matters to a lesser degree.

To shed light on the feasibility of strategies that hedge out unpriced risks, we provide sensitivity analyses and impose restrictions on the absolute values of portfolio weights. We document that portfolio performance is remarkably robust to imposing such limits. Only when long-only restrictions are imposed, the portfolio performance drops. However, bond futures exist in almost all of the G10* markets, which should make shorting a particular market easy for professional market participants.

Recognizing that the existing finance literature provides little theoretical and empirical foundation on which international government bond portfolio strategies can be built, this paper provides a first step to fill this gap. However, given the growing size and economic importance of this asset class, this is likely to remain an active area of research. One obvious extension is to analyze pricing kernels for a broader universe of bond markets which also includes emerging markets, where default risks are likely to play a role.

CRedit authorship contribution statement

Otto Randl: Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Conceptualization. **Giorgia Simion:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Josef Zechner:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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³⁰ For details, see Equation (17) from Sandulescu et al. (2021).

Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jfineco.2025.104152>.

Data availability

Pricing and Constructing International Bond Portfolios (Original data) (Mendeley Data)

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